

Meritocratic Selection and Dynamics of Coordination Neglect in Hierarchical Organizations*

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We study how meritocratic promotion affects coordination neglect in hierarchical organizations. In an overlapping-generations model, managers and subordinates allocate effort between independent and coordination tasks, with the top-performing subordinate being promoted. Individuals undervalue coordination and expect others to share their biases. The long-run effect of meritocracy depends on whether coordination efforts are substitutes or complements. When they are substitutes, neglect is detrimental, and meritocratic selection reduces bias, improving performance. When they are complements, performance peaks at moderate bias; in large organizations, selection drives managerial bias below the optimal level, making random promotion more effective than meritocracy.

JEL Classification: D82, D91, M51

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1. Introduction

Coordination among members shapes how well organizations perform, but workers routinely underestimate how much it matters. Evidence from Google’s Project Aristotle makes the point concretely: the collaborative activities most strongly associated with team success are precisely those that team members themselves consistently fail to recognize as important ([Duhigg, 2016](#)). [Heath and Staudenmayer](#)

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(2000) trace this pattern to a general cognitive tendency, which they call *coordination neglect*: individuals systematically discount the value of coordinating with others and act accordingly.

What drives coordination neglect is, however, less settled. Heath and Staudenmayer (2000) locate the source of this tendency in cognition: workers hold an intuitive view of work in which individual effort is the primary input, leading them to discount the value of coordination independent of how it is rewarded. An alternative account attributes underinvestment in coordination primarily to institutional factors: when performance metrics focus on individual output, workers have little reason to invest in coordination, and underinvestment follows as a rational response (e.g., Malone and Crowston, 1994; Castilla, 2025). If this alternative account is correct, redesigning the evaluation system should be sufficient to correct the problem; if the source is instead cognitive, such a redesign may leave the underlying bias intact.

This raises a question for organizational design. Even if coordination neglect is a cognitive bias that cannot be eliminated by redesigning incentives, one might still hope that meritocratic selection can correct it over time: by consistently favoring less biased workers for promotion, a metric that rewards coordination alongside individual output should gradually screen out the more biased individuals from managerial positions. This paper investigates whether that intuition holds, and shows that the results depend on a single technological feature of the environment: whether the coordination efforts of the manager and subordinates are *substitutes* or *complements*, with sharply contrasting implications for long-run organizational performance.

We study an overlapping-generations organization in which a manager and a pool of subordinates allocate effort between independent and coordination tasks. Individuals differ in their degree of coordination-neglect bias, modeled as a misperception that scales down the perceived value of coordination relative to independent work. In each period, the subordinate with the highest real performance is promoted, measured by a comprehensive metric that rewards both types of output. Although subordinates are drawn from a population with a fixed distribution of bias in each generation, the distribution of managerial bias evolves endogenously through selection.

A central feature of our model is that coordination neglect is naïve: each individual expects others to share her degree of bias.¹ Because the bias is cognitive, an individual who undervalues coordination perceives her evaluation as correct and expects others to share it.² As a result, a less biased individual anticipates that her partner will invest more in coordination than a more biased individual, which in turn affects effort responses and performance outcomes in ways that critically depend on technology:

¹We use female pronouns for the manager and for a generic individual or player, and male pronouns for the subordinate.

²This naïveté assumption is related to the literature on information projection bias; see the Related Literature section for further discussion.

whether the coordination efforts are substitutes or complements. This distinction leads to two sharply contrasting results.

When coordination efforts are substitutes, meritocratic selection reduces coordination neglect and improves long-run performance. Bias is unambiguously harmful to performance; less biased subordinates consistently outperform more biased ones; and selection favors low-bias managers. The managerial bias distribution is first-order stochastically dominated by the original subordinate pool in every generation, so long-run expected performance exceeds what random promotion would deliver. This is the regime in which meritocratic logic is self-consistent: rewarding coordination selects for coordination, and the organization corrects its own bias over time.

When coordination efforts are complements, the same meritocratic institution produces fundamentally different dynamics. A subordinate now faces a tradeoff between two pressures. On the one hand, complementarity favors alignment with the manager: when biases match, the two parties' coordination efforts complement each other most effectively. On the other hand, the metric rewards the coordination effort that the output actually requires, not the effort that either party perceives as optimal. The resolution is a subordinate whose bias lies slightly below the manager's: biased enough to align with the manager's coordination effort, but unbiased enough to perform well on the metric. Subordinate performance is therefore single-peaked in own bias, with a peak strictly below the manager's level, and meritocratic selection promotes moderately biased subordinates rather than the least biased ones.

This single-peakedness of subordinate performance shapes how selection operates, but the direction of selection depends on where the manager's bias lies. The key distinction is whether complementarity is moderate or strong, determined by whether an unbiased subordinate can outperform the most biased one under a maximally biased manager. If so, complementarity is moderate, and selection never amplifies bias: depending on the manager's bias level, the next generation's distribution is either shifted toward lower bias or compressed without a clear directional shift. If not, complementarity is strong, and a richer non-monotone pattern emerges. Selection reduces bias under less biased managers, compresses its dispersion at intermediate levels, and amplifies bias under highly biased managers, shifting the next generation toward higher bias in the FOSD order.

The implications become more striking in large organizations. When the subordinate pool is large, a subordinate whose bias is close to the performance-maximizing level is almost certain to exist, and selection picks him with probability approaching one. Managerial bias therefore decreases monotonically across generations and, in the limit, collapses to the minimum of the subordinate distribution. But overall performance is single-peaked in managerial bias, with a peak at a strictly positive interior level:

some coordination neglect is beneficial because it prevents the manager from overinvesting in coordination that biased subordinates cannot reciprocate. Meritocracy therefore overshoots: by eliminating bias beyond the performance-maximizing level, it selects managers who are too cooperative for their subordinates to match, and long-run performance declines.³ We show that random promotion, by preserving diversity in managerial bias, strictly dominates performance-based promotion in the large-organization limit.

A well-designed meritocracy that explicitly rewards coordination output alongside individual output reliably reduces coordination neglect when efforts are substitutes, but may actively work against effective coordination when efforts are complements. The technology of coordination, not the design of the metric, is the binding constraint. This suggests a constructive response: in complement environments, promotion schemes that deliberately preserve diversity in managerial coordination styles, through rotation, quotas, or hybrid rules that combine performance with random selection, can outperform pure meritocracy because they maintain precisely the diversity that pure meritocracy tends to eliminate.

This paper makes two related contributions to the literature on organizational behavior and promotion design. First, we provide a dynamic framework, built on first-order stochastic dominance comparisons across heterogeneous-bias profiles, for studying how organizational coordination culture evolves under performance-based promotion. Second, we identify the technological structure of coordination as a primary determinant of whether meritocratic institutions help or hurt, and we show that in the complements case meritocracy faces an internal tension: the metric that rewards coordination also selects for a degree of coordination neglect that misaligns managerial and subordinate effort. These findings have direct implications for evaluation and promotion design in coordination-intensive hierarchical organizations, where the substitutes-complements divide implies sharply different optimal policies.⁴

Related Literature

Our paper primarily contributes to the literature on the evolution of psychological biases. While most prior studies have focused on overconfidence (e.g., [Johnson and Fowler, 2011](#); [Bernardo and Welch, 2001](#); [Goel and Thakor, 2008](#); [Heller, 2014](#)),⁵ our work is, to our knowledge, the first to analyze how

³Although this result is established in the large-organization limit, our numerical analysis in the supplementary material shows that performance declines arise at moderate team sizes: at $n = 25$ and ten promotion rounds, a substantial fraction of initial bias distributions already produce negative changes in long-run performance.

⁴Section 2.1 provides concrete examples on both sides of this divide.

⁵[Johnson and Fowler \(2011\)](#) provide evolutionary dynamics that enable overconfident individuals to survive in several strategic environments. [Bernardo and Welch \(2001\)](#) analyze how an overconfident individual with private information is promoted in a situation where an information cascade arises as a problem. [Goel and Thakor \(2008\)](#) rationalize the selection of an overconfident manager as a CEO in an environment where risk-taking behavior results in promotion. [Heller \(2014\)](#) shows that a risk-averse principal may prefer privately informed overconfident agents in a strategic situation and presents an

coordination neglect evolves through selection and strategic interaction in hierarchical organizations. Our model highlights a unique asymmetry: while subordinates are drawn from a fixed bias distribution, managers are selected based on endogenous performance outcomes. This allows us to study the evolution of bias through a stochastic dominance framework and uncover unintuitive consequences for organizational performance.

Our model can be broadly viewed as a variant of multitask (e.g., [Holmstrom and Milgrom, 1991](#)) and multi-agent or team-production (e.g., [Holmstrom, 1982](#)) frameworks. In a seminal paper incorporating both features, [Itoh \(1991\)](#) analyzes a multi-agent environment where each worker allocates effort between independent tasks and provides “help” to others, and characterizes the incentive schemes that induce either specialization or teamwork.⁶ These studies examine moral-hazard environments in which certain task outcomes are unverifiable and aim to create incentives for effort on such tasks. In contrast, our setting does not exhibit preference misalignment and information asymmetry; all outputs, including coordination efforts, are verifiable and contribute to the performance metric. We instead examine whether rewarding coordination is sufficient to encourage coordination effort and how managerial coordination neglect evolves through meritocratic promotion, ultimately influencing long-run organizational performance.

Our analysis also relates to the literature on information projection bias (e.g., [Madarász, 2012](#); [Madarász, Danz, and Wang, 2024](#)).⁷ While our formulation interprets the parameter as capturing the extent to which workers undervalue coordination tasks as a form of coordination neglect, it can also be viewed as a distortion in beliefs about others’ information states or cooperative intentions. In settings where communication is costly (e.g., [Dewatripont and Tirole, 2005](#); [Fehr, 2017](#)), information projection bias reduces the perceived value of information exchange, leading individuals to underestimate the benefits of coordination. As a result, people underinvest in communication and coordination ([Madarász, 2012](#)). This mechanism is analogous to the behavioral friction we study: individuals expect others to coordinate “too much like themselves,” causing them to underestimate the benefits of coordination and collaborative efforts. By embedding such a distortion into a dynamic organizational environment with meritocratic promotion, our model highlights how biased perceptions of cooperation can either persist or diminish

evolutionary foundation for overconfidence. Other than overconfidence, [Heller and Nehama \(2023\)](#) examine the evolution of risk attitudes among agents who face the same distribution of risk preferences.

⁶[Ishihara \(2017\)](#) extends the framework of [Itoh \(1991\)](#) to a relational contracting setting.

⁷In relation to projection bias, [Gagnon-Bartsch \(2016\)](#), [Gagnon-Bartsch, Pagnozzi, and Rosato \(2021\)](#), and [Gagnon-Bartsch and Rosato \(2024\)](#) examine taste projection. [Bushong and Gagnon-Bartsch \(2024\)](#) demonstrate that interpersonal taste projection bias results in significant and costly errors when individuals forecast others’ behavior in a real-effort experiment. [Morita, Muramoto, and Sogo \(2025\)](#) investigate optimal contracts within a moral hazard framework where multiple agents exhibit projection bias.

across generations, and how institutional mechanisms interact with these frictions to influence long-run organizational performance.

Organization of the Article

The remainder of the paper is organized as follows. Section 2 introduces our model: Section 2.1 illustrates the motivating environments; Section 2.2 defines the formal OLG organizational model; and Section 2.3 defines coordination-neglect bias and behavioral equilibrium under heterogeneous bias profiles. After analyzing the benchmark case without bias, Section 3 characterizes equilibrium effort choices when coordination neglect is present. Section 4 studies the case in which cooperative efforts are technologically substitutable, characterizing equilibrium properties, bias dynamics, and implications for long-run performance. Section 5 turns to the complementary-effort case, establishing the single-peakedness of performance, deriving the dynamics of bias evolution, and showing how meritocratic promotion can ultimately reduce long-run performance in large organizations. Section 6 examines the robustness of the main results under two extensions of the baseline model and discusses the empirical implications of our findings.

2. Model

2.1. Motivating Environments and Examples

As realistic work environments captured by our model, we envision apprenticeship-like, coaching-intensive organizations characterized by internal promotion and manager-held expertise.⁸ In such environments, managers are responsible not only for their own output but also for organizing and supporting their subordinates' work. Subordinates, in turn, divide their time between producing individual deliverables and engaging in activities that facilitate effective coordination with the manager. The following examples illustrate these environments.⁹

Example 1. In many professional service organizations, such as consulting, law, and accounting, a partner supervises several associates. Both parties allocate time between independent tasks, such as drafting analyses, writing legal documents, or creating slide decks, and coordination tasks, which involve developing a coherent client strategy, aligning on key arguments, or preparing integrated deliverables. The partner's coordination efforts include organizing project kick-off and status meetings, providing feed-

⁸Here, 'manager-held expertise' should be interpreted as role-specific coordinating capacity, such as authority, access to information, and accumulated experience, rather than a skill that appears discontinuously upon promotion.

⁹We thank an anonymous reviewer for the suggestion of these concrete examples.

back, and integrating different components of the work. Associates must, in turn, prepare for these meetings, share information, and adjust their work in accordance with guidance.

Example 1 illustrates the substitutes case, in which coordination efforts are substitutes. An associate who anticipates limited guidance from the partner can compensate by spending more time refining their own drafts or over-preparing their individual sections. Likewise, when partners expect associates to “figure things out on their own,” they may reduce meeting time and direct supervision, relying on associates to work more independently.

Example 2. In cross-functional product development or software engineering teams, engineers implement separate modules, but system performance depends critically on how well these modules interface. The manager organizes design meetings, maintains shared documentation, coordinates integration tests, and resolves cross-team interdependencies. Engineers must invest time in understanding the architecture, participating in design reviews, and adapting their code based on feedback and the requirements of other modules.

Example 2 illustrates the complements case. The value of an engineer’s coordination efforts, such as time spent on integration, documentation, or joint debugging, is higher when the manager also invests in architecture discussions, coordination meetings, and integration planning. Conversely, the manager’s coordination efforts are productive only if engineers dedicate sufficient time to them. When either side neglects coordination, misalignment increases, and system failures or costly rework become more likely.

Example 3. Teaching hospitals require substantial coordination between attending physicians and residents. Attending physicians supervise residents on the ward, balancing their time between independent tasks, such as direct patient care, and coordinating activities, including managing the care team through ward rounds, case conferences, and teaching. Residents must allocate their time between documentation, tests, and procedures, while also participating in rounds, sharing information, and aligning their actions with the attending physician’s plan.

Example 3 also illustrates the complements case: effective patient care depends on the guidance of the attending physician and the responsiveness of residents.

All examples describe environments in which coordination-neglect bias can arise, as individuals may underestimate the value of coordination relative to more visible individual contributions. A partner who undervalues coordination may schedule fewer meetings and invest less in feedback, assuming that associates can effectively self-organize. An engineer who neglects coordination may focus solely on their own module, leading to insufficient investment in integration or documentation. Similarly, a resident

who underappreciates the importance of rounds may prioritize independent tasks over team discussions.

The three examples differ not only in whether coordination efforts are substitutes or complements but also in the typical scale of the teams involved. In Example 1, team size varies significantly: small consulting or law practices may have only a few associates per partner, while larger firms often form project teams comprising dozens or even hundreds. In Example 2, team size also varies widely, ranging from small startups with a few engineers to large technology organizations with many engineers per manager. In contrast, Example 3 is inherently small-scale; typically, a single attending physician supervises three to six residents, representing a situation with limited competition among team members.

2.2. Primitives

Consider an OLG organization consisting of a manager and n subordinates at each period t .¹⁰ We denote the manager as m and a typical subordinate as s . Each player can engage in both independent and coordination tasks. The outcome of each independent task depends solely on each player's effort. In contrast, a coordination task depends on the efforts of the subordinate and the manager. For example, as described in Section 2.1, coordination tasks capture the manager's coordination and communication activities that allow individual contributors to effectively combine their work, as well as subordinates' efforts to adapt their work in response to the manager's guidance.

Each player must decide how to allocate their efforts between independent and coordination tasks. We represent each player's effort allocation as $(e_k, 1 - e_k)$ where $e_k \in [0, 1]$ and $k \in \{m, s\}$. e_k denotes the effort for the independent task, with its outcome represented by $u(e_k)$, while $1 - e_k$ denotes the effort devoted to the coordination task, with its outcome represented by $v(e_s, e_m)$.¹¹ We study two cases: coordination efforts are substitutes when $v_{12} < 0$ and complements when $v_{12} > 0$, where $v_{12} \equiv \frac{\partial^2 v(e_s, e_m)}{\partial e_s \partial e_m}$.¹²

We also assume the following properties, which guarantee the concavity of the objective functions and clean comparative statics.

¹⁰We omit t to economize on notation when it is clear from context.

¹¹Managerial efforts for coordination can be further divided into two types. Some activities benefit all subordinates simultaneously, such as preparing and distributing common project documents, holding kick-off meetings, or organizing training sessions. Other activities are targeted at individual subordinates, such as one-on-one project reviews or tailored coaching. To capture this distinction in a simple way, we assume that the value of the coordination task for each subordinate increases in $(1 - e_m)(\varrho + (1 - \varrho)\frac{1}{n})$, where $1 - e_m$ is the manager's total effort devoted to coordination tasks and $\varrho \in [0, 1]$ measures the share of "collective" effort. When ϱ is high, most coordination effort takes the form of activities that benefit all subordinates at once; when ϱ is low, the manager's effort mainly consists of one-on-one interactions that are effectively diluted across n subordinates.

As we show in the following sections, our qualitative results are unchanged even if v depends on the number of subordinates n , as long as $\varrho > 0$. Intuitively, a positive share of collective managerial effort ensures that the marginal effect of coordination effort on each subordinate's performance does not vanish as the organization grows, so the comparative statics with respect to the bias parameter and the nature of coordination remain the same.

¹²Hereafter, v_i denotes the partial derivative of v with respect to its i -th argument; similarly, v_{ij} denotes the partial derivative of v_i with respect to its j -th argument.

Assumption 1. (i) For each $i \in \{1, 2\}$, $v_{ii} < 0$ and $u'' < 0$.¹³

(ii) $(u''(e))^2 \geq (v_{12}(e_s, e_m))^2$ for any $e_s, e_m, e \in [0, 1]$.

(iii) For any $e_m, e_s \in [0, 1]$, $\min\{v_1(1, e_m), v_2(e_s, 1)\} > -u'(1) > -u'(0) > \max\{v_1(0, e_m), v_2(e_s, 0)\}$.

(iv) For any $e_m, e_s, e \in [0, 1]$, $u''(e) < \max\left\{\frac{v_1(e_s, e_m)}{v_2(e_s, e_m)}, \frac{v_2(e_s, e_m)}{v_1(e_s, e_m)}\right\} v_{12}(e_s, e_m)$.

(v) $v_{12} > 0$ or $v_{12} < 0$.

Assumption 1 (i) and (ii) guarantee the concavity of the objective functions. Assumption 1 (v) states that the coordination efforts are either complements ($v_{12} > 0$) or substitutes ($v_{12} < 0$), and that this relationship is fixed over the relevant range. Assumption 1 (iii) and (iv) are mainly technical and ensure monotone comparative statics.¹⁴ As an example, the following quadratic functions satisfy the assumption for sufficiently large ζ .

$$\begin{aligned} u(e) &= \zeta(e - e^2/2), \\ v(e_s, e_m) &= \beta_s(1 - e_s) - \delta_s(1 - e_s)^2/2 + \gamma(1 - e_m)(1 - e_s) + \beta_m(1 - e_m) - \delta_m(1 - e_m)^2/2, \\ \zeta &> 0, \quad \beta_k > 0, \text{ and } \delta_k > 0 \text{ for each } k \in \{s, m\}. \end{aligned} \quad (1)$$

In this specification, v is complements when $\gamma > 0$ and substitutes when $\gamma < 0$.

The manager at t is selected from the subordinates at $t - 1$. We refer to the promotion system as *performance-based* if the subordinate with the highest performance among the subordinates is promoted to the next-generation manager position. The performance of subordinate s depends on the performance shock, the coordination task, and the independent task, which is represented by $\pi_s = \theta_s + v(e_s, e_m) + u(e_s)$.¹⁵ Importantly, promotions are based on a performance metric that reflects coordination as well as independent outcomes. The term θ_s represents the performance shock of subordinate s , which is identically and independently distributed by the cumulative distribution function (cdf) F on $\mathbb{R}$ for each subordinate. The average is denoted by $\bar{\theta}$. We assume that $\lim_{x \rightarrow \infty} \frac{f(x-a)}{f(x-b)} = 0$ for each $b > a$.¹⁶ The organization's overall performance is given by $\Pi = u(e_m) + \sum_{j \in N} \frac{\pi_j}{n}$.¹⁷

¹³For any function $\varphi: X \rightarrow Y$, $\varphi > 0$ indicates that $\varphi(x) > 0$ for any $x \in X$.

¹⁴The role of Assumption 1 (iii) is explained below equation (2) in Section 3, while the role of Assumption 1 (iv) is discussed in Remark 1 of Section 4.1.

¹⁵One implication of θ may include the ability of the subordinate. In this context, alternative formulations related to performance can be considered, such as $\theta v + u$ and $v + \theta u$. However, these formulations implicitly assume a correlation between ability and effort allocation. If we instead assume $\theta(v + u)$, θ does not influence the decision-making process under the assumption that effort is costless. Therefore, to simplify the dynamics, we assume that θ and the task values are independent.

¹⁶For example, normal distributions satisfy this assumption.

¹⁷This specification normalizes the performance metric so that the output of the manager's independent task, $u(e_m)$, is exactly

We assume that each subordinate maximizes his performance π_i ,¹⁸ while the manager maximizes the overall performance Π . Finally, the timing is as follows:

1. Each player simultaneously determines their effort allocation.
2. Each performance is revealed, and the promotion is decided.

2.3. Coordination Neglect Bias and Definition of Equilibrium

This study incorporates the behavioral tendency of organizational members to underestimate the importance of coordination. Following [Heath and Staudenmayer \(2000\)](#), we refer to this tendency as coordination neglect. We interpret coordination neglect as a misperception of the payoff from coordination rather than as preference heterogeneity. We denote the degree of this coordination-neglect bias as $\beta \in [0, 1]$. A player with bias β evaluates the coordination task as $(1 - \beta)v(e_s, e_m)$ instead of $v(e_s, e_m)$, and is naive about this distortion: she takes her own evaluation as the correct one and therefore expects others to assess the coordination task in the same way. Formally, each player believes that all players share her β , even though in reality different players may differ in their degrees of coordination neglect. This naïveté assumption allows us to abstract from higher-order belief considerations.¹⁹

To analyze this situation, given the degree of bias β , we first construct a ‘subjective game’ in which both the manager and all subordinates evaluate the coordination task with β and simultaneously choose their efforts. We then use the best responses to analyze heterogeneous bias profiles.

In this subjective game, a subordinate s chooses e_s to maximize

$$\pi_s = \theta_s + (1 - \beta)v(e_s, e_m) + u(e_s),$$

and the manager chooses e_m to maximize

$$u(e_m) + \frac{1}{n} \sum_{j \in N} \pi_j.$$

Let (e_s^β, e_m^β) denote the unique Nash equilibrium of this subjective game.

n times as important as each subordinate’s performance π_j . This normalization removes a purely mechanical dependence of equilibrium effort levels on team size n : without it, increasing n would automatically raise the weight on subordinates’ output relative to the manager’s, even if the underlying technology were unchanged. The relative importance of the manager’s task versus the coordination technology is instead captured by the curvature of v and u , so the qualitative results are invariant to this scaling.

¹⁸This is a reduced-form representation of promotion incentives and performance-based bonuses. We will validate the robustness of the promotion incentives in Section 3.

¹⁹We will discuss its cognitive foundations and scope at the end of this subsection.

In the organization described in our model, a subordinate with the degree of bias $\alpha \in [0, 1]$ behaves as if the true game were the subjective game with bias $\beta = \alpha$, and therefore chooses e_s^α , taking e_m^α as the manager's best response. Similarly, a manager with the degree of bias $\hat{\alpha} \in [0, 1]$ behaves as if the game were the subjective game with bias $\beta = \hat{\alpha}$ and chooses $e_m^{\hat{\alpha}}$, taking $e_s^{\hat{\alpha}}$ as the subordinate's best response. Hence, under the bias profile $(\alpha, \hat{\alpha})$, the realized effort profile is $(e_s^\alpha, e_m^{\hat{\alpha}})$. We refer to this profile as a behavioral equilibrium with heterogeneous degrees of coordination-neglect bias.

Let G denote the cross-sectional distribution of subordinates' degrees of bias, which is assumed to be fixed over generations, and let $\hat{G}$ denote the distribution of managers' degrees of bias, which evolves endogenously through the promotion process. For notational simplicity, we write α for the typical subordinate's degree of bias and $\hat{\alpha}$ for the manager's degree of bias.

We emphasize that the naïveté assumption concerns only the beliefs each party holds about the other in the hierarchical relationship: subordinates' beliefs about the manager and the manager's beliefs about subordinates—not subordinates' beliefs about one another. In our model, each subordinate's performance π_i depends on his own effort and the manager's effort, but not on the efforts or biases of fellow subordinates. Subordinates interact only indirectly, through the promotion contest that compares realized performances.

A related question is whether the naïveté assumption holds in small teams, where subordinates can easily observe each other's actions. One might expect that observing a colleague invest heavily in coordination would lead a subordinate to revise upward his valuation of coordination. However, we argue that such updating is unlikely because the naïveté assumption operates at the level of subjective payoff evaluation rather than beliefs about observable actions. A subordinate who undervalues coordination may observe a colleague devoting time to meetings or integration activities but interpret this as excessive or unproductive effort, rather than as evidence that coordination is more valuable than he perceives. This reflects the cognitive structure identified by [Heath and Staudenmayer \(2000\)](#): coordination neglect arises from mental models that are cognitively entrenched and not easily revised through casual observation of peers. We therefore treat naïveté as exogenous, based on this cognitive structure. This aligns with our model, where subordinates have no direct strategic reason to infer the bias levels of other subordinates.

3. Equilibrium with Coordination-Neglect Bias

As a benchmark, we first analyze the case where players do not have the bias, in which case the effort allocation that maximizes total performance, $u(e_m) + v(e_s, e_m) + u(e_s)$, satisfies the following first-order

conditions with respect to e_s and e_m .

$$v_1(e_s, e_m) + u'(e_s) = 0,$$

$$u'(e_m) + v_2(e_s, e_m) = 0.$$

These conditions are the same as those in equilibrium, as the manager's objective is $u(e_m) + v(e_s, e_m)$, and the subordinate's objective is $u(e_s) + v(e_s, e_m)$. Therefore, equilibrium is efficient when the players do not exhibit coordination neglect. The following is a summary.

Proposition 1. *When no player exhibits coordination neglect, the equilibrium effort profile maximizes the overall performance.*

We now examine the case in which players exhibit coordination-neglect bias. By comparing the results affected by coordination neglect with those in Proposition 1, we can clarify how this bias leads to inefficiency and whether the performance-based promotion system can alleviate it.

When a player, either a subordinate or a manager, has bias β , the player's perceived optimal effort level, (e_s^β, e_m^β) , is given as a solution to the following system of inequalities, which are the first-order conditions allowing for the possibility of corner solutions, where $e_k = 1$ for each $k \in \{s, m\}$.

$$\begin{aligned} u'(e_s^\beta) + (1 - \beta)v_1(e_s^\beta, e_m^\beta) &\geq 0 \quad \text{equality holds if } e_s^\beta < 1, \\ u'(e_m^\beta) + (1 - \beta)v_2(e_s^\beta, e_m^\beta) &\geq 0 \quad \text{equality holds if } e_m^\beta < 1. \end{aligned} \tag{2}$$

We exclude the corner solution when the player has bias $\beta = 0$ for simplicity, which is implied by Assumption 1 (iii). We obtain the following result by showing that the solutions to (2) uniquely exist.

Proposition 2. *The subjective game has a unique equilibrium for any $\beta \in [0, 1]$. For any $\alpha, \hat{\alpha}$, $e_s^\alpha > 0$ and $e_m^{\hat{\alpha}} > 0$, and $e_s^0 < 1$ and $e_m^0 < 1$.*

Hereafter, we denote by $(e_s^\alpha, e_m^{\hat{\alpha}})$ the unique equilibrium effort profile under the bias profile $(\alpha, \hat{\alpha})$.

Robustness to Promotion Incentives In our baseline model, each subordinate is assumed to maximize his own performance π_i , which we interpret as a reduced form for promotion incentives and performance-based compensation. A natural concern is whether explicitly incorporating the value of promotion into subordinates' objectives would alter the equilibrium effort choices. We now show that, under a mild monotonicity condition on the promotion rule, the answer is no: explicitly modeling promotion incentives leaves the equilibrium effort profile unchanged.

Consider an extension of the baseline model in which subordinate i 's payoff is augmented by a promotion incentive. Let $M_i((\pi_j)_{j \in N})$ denote the probability that subordinate i is promoted, given the realized performance profile $(\pi_j)_{j \in N}$. Under the performance-based promotion rule of our model, M_i is the probability that π_i is the highest among all subordinates, but our argument applies to any promotion rule for which the following monotonicity condition holds:

$$\frac{\partial M_i}{\partial \pi_i} \geq 0. \quad (3)$$

This condition states that raising one's own performance never decreases the probability of being promoted. It is satisfied by virtually any reasonable promotion rule, including the top-performer rule of our baseline, rank-based rules, and Tullock-style probabilistic rules.

Subordinate i 's augmented payoff is $\pi_i + \rho M_i((\pi_j)_{j \in N})$, where $\rho > 0$ measures the value of promotion (e.g., the size of the promotion prize or the expected wage gain associated with promotion).

Lemma 1. *Consider the extended model in which each subordinate maximizes $\pi_i + \rho M_i((\pi_j)_{j \in N})$, where M_i satisfies (3). Then, for every $\rho \geq 0$, the equilibrium effort profile coincides with the baseline equilibrium $(e_s^\alpha, e_m^{\hat{\alpha}})$.*

Proof. Since e_i enters subordinate i 's augmented payoff only through π_i , the first-order condition for e_i takes the form $(1 + \rho \partial M_i / \partial \pi_i) \partial \pi_i / \partial e_i = 0$. The factor in parentheses is strictly positive by (3) and $\rho \geq 0$, so the FOC reduces to $\partial \pi_i / \partial e_i = 0$, which coincides with the baseline subjective game. The manager's problem is unaffected by ρ . $\square$

Notably, this argument requires neither symmetry across subordinates nor any assumption about how subordinates form beliefs about their competitors. It relies only on the fact that one's own effort affects only one's own performance, combined with the monotonicity of the promotion rule.

4. The Substitutes Case

This section examines the case in which coordination efforts are substitutes. The performance of subordinate i is $\theta_i + v(e_s^\alpha, e_m^{\hat{\alpha}}) + u(e_s^\alpha)$ when biases of the manager and subordinate are given. Hereafter, we consider the performance attributed to efforts, $x^{\hat{\alpha}}(\alpha) = u(e_s^\alpha) + v(e_s^\alpha, e_m^{\hat{\alpha}})$, to focus on the impact of bias on performance. We also define $E[\Pi \mid \hat{\alpha}] = u(e_m^{\hat{\alpha}}) + \int x^{\hat{\alpha}}(\alpha) dG(\alpha)$ as the expected overall performance when the manager's bias is $\hat{\alpha}$.

4.1. Equilibrium Properties

The following lemma characterizes the equilibrium properties between bias degrees, effort allocations, and performances. The results are used to characterize bias dynamics and induce welfare implications.

Lemma 2. *Suppose that $v_{12} < 0$. Then, we have the following properties:*

- (a) $\frac{\partial e_m^{\hat{\alpha}}}{\partial \hat{\alpha}} \geq 0$ and $\frac{\partial e_s^{\alpha}}{\partial \alpha} \geq 0$. The inequality is strict if $e_m^{\hat{\alpha}} < 1$ and $e_s^{\alpha} < 1$.
- (b) $x^{\hat{\alpha}}(\alpha)$ is non-increasing in $\hat{\alpha}$ and α , and strictly decreasing if $e_m^{\hat{\alpha}} < 1$ and $e_s^{\alpha} < 1$.
- (c) $\frac{\partial^2 x^{\hat{\alpha}}(\alpha)}{\partial \alpha \partial \hat{\alpha}} \leq 0$. The inequality is strict if $e_m^{\hat{\alpha}} < 1$ and $e_s^{\alpha} < 1$.
- (d) $E[\Pi \mid \hat{\alpha}]$ is non-increasing with respect to $\hat{\alpha}$, and strictly decreasing if $e_m^{\hat{\alpha}} < 1$.

The implications of Lemma 2 are as follows. First, Lemma 2 (a) shows that, with a higher degree of bias, players exert less coordination effort. This is intuitive because coordination neglect leads players to perceive a lower need for coordination effort.

Lemma 2 (b)–(c) concern the subordinate's performance. Specifically, (b) shows that subordinate performance declines as the degrees of bias of both the manager and the subordinate increase. Lemma 2 (c) captures the implications of the substitutes case: when the manager is more biased and thus provides less coordination effort, the marginal effect of subordinate's bias on performance becomes more negative.

Finally, Lemma 2 (d) shows that the overall expected performance decreases with managerial bias. This is because bias reduces subordinates' coordination effort. Importantly, the best performance is achieved when neither party has bias, as this maximizes the actual overall performance on both sides (Proposition 1). Since coordination effort are otherwise below the efficient level, greater managerial effort can partially compensate and thereby improve overall performance through substitutability.

Remark 1. Lemma 2 (a) relies on Assumption 1 (iv).²⁰ This condition restricts the degree of substitutability to avoid excessive strength. Intuitively, it bounds the cross effect v_{12} relative to the own curvatures of u and v , ensuring that optimal coordination effort are monotone in the bias parameter α and yielding clean comparative statics.

4.2. Dynamics of Bias and Performance

We analyze the dynamics of coordination-neglect bias in the substitutes case. Suppose the manager's bias $\hat{\alpha}$ is distributed according to $\hat{G}$, with density $\hat{g}$. We denote $\mathcal{G}_{\hat{G}, G}$ as the distribution of the bias

²⁰Note that this assumption imposes no restriction when $v_{12} > 0$, in which case Assumption 1 (iv) is implied by the other assumptions (namely, $v_1 < 0$, $v_2 < 0$ and $u'' < 0$).

of the best-performing subordinate, conditional on the manager's bias following $\hat{G}$ and the subordinates' biases following G .²¹ By abusing notation, we use $\mathcal{G}_{\hat{\alpha},G}$ to denote the distribution of the best-performing subordinate's bias, conditional on the manager's bias being fixed to $\hat{\alpha}$. Therefore, $\mathcal{G}_{\hat{G},G}(\alpha) = \int \mathcal{G}_{\hat{\alpha},G}(\alpha) \hat{g}(\hat{\alpha}) d\hat{\alpha}$.

Specifically, if $\hat{G}_t$ is the current manager's bias distribution, the next-generation manager's bias distribution is $\hat{G}_{t+1} = \mathcal{G}_{\hat{G}_t,G}$. We assume that $\hat{G}_0 = G$. Let $>_{\text{fsd}}$ represent the (strict) *first-order stochastic dominance* (FOSD) relation. Also, let $\succeq_{\text{fsd}}$ be the weak dominance relation.

By exploiting the properties shown in Lemma 2, we establish the following characterization of the evolution of the managerial bias distribution.

Theorem 1. *Suppose that $v_{12} < 0$. The following statements hold for any distribution G :*

- (a) *For any distribution functions $\hat{G}$ and $\bar{G}$, if $\hat{G} >_{\text{fsd}} \bar{G}$, $\mathcal{G}_{\bar{G},G} >_{\text{fsd}} \mathcal{G}_{\hat{G},G}$.*
- (b) *For any distribution $\hat{G}$, $G >_{\text{fsd}} \mathcal{G}_{\hat{G},G}$.*
- (c) *Consider the dynamics of manager's bias distribution $(\hat{G}_t)_{t=0}^{\infty}$ with $\hat{G}_0 = G$. Then, for each $t \geq 2$,*

$$\begin{aligned} G &>_{\text{fsd}} \hat{G}_{t-2} >_{\text{fsd}} \hat{G}_t >_{\text{fsd}} \hat{G}_{t-1} >_{\text{fsd}} \hat{G}_1 && \text{if } t \text{ is even,} \\ G &>_{\text{fsd}} \hat{G}_{t-1} >_{\text{fsd}} \hat{G}_t >_{\text{fsd}} \hat{G}_{t-2} >_{\text{fsd}} \hat{G}_1 && \text{if } t \text{ is odd.} \end{aligned}$$

- (d) *$\hat{G}_t$ converges to a limit distribution $\hat{G}$.*

Theorem 1 (a) indicates a selection effect: managers with higher bias tend to promote subordinates with lower bias, leading to a downward shift in bias across generations. The intuition behind this selection effect is provided by Lemma 2 (c), which shows that, in the substitutes case, a less biased subordinate performs better when matched with a highly biased manager. This result follows from the substitutability of coordination efforts: lower coordination effort by the manager raises the subordinate's optimal coordination effort.

Although Theorem 1 (a) is a comparison between the managers' bias distributions, Theorem 1 (b) compares the bias distributions between subordinates and *selected* subordinates. It shows that, under the promotion system, the selected subordinates, who become the next managers, tend to be less biased than other subordinates. This property comes from Lemma 2 (b), which shows that subordinates with lower bias yield higher performance.

²¹The formal derivation is relegated to Appendix A.

Building on properties shown by Theorem 1 (a) and (b), Theorem 1 (c) characterizes the entire evolution dynamics in terms of stochastic dominance. It shows that, although the sequence does not improve monotonically period by period, it converges in the sense of FOSD. This dynamic reflects an alternating pattern: even-numbered generations become progressively less biased, and the odd-numbered generations goes the opposite direction, leading to bias distributions that are consistently less biased than the subordinates' distribution in the sense of FOSD.

As Lemma 2 (d) shows, since $E[\Pi \mid \hat{\alpha}]$ is decreasing in $\hat{\alpha}$, the overall productivity is negatively correlated with the degree of managerial bias. Therefore, although overall productivity fluctuates over time, it tends to remain higher than the initial level, as managers' biases are consistently lower than at the beginning. In this sense, the performance-based promotion system outperforms alternative systems that simply preserve subordinates' bias distributions, such as random promotion. However, managerial bias, while detrimental to overall performance, is not eliminated through the promotion process.

We remark that Lemma 2 (b) implies that $x^{\hat{\alpha}}(\alpha)$ is decreasing in α . In other words, a subordinate with $\alpha = \min \text{supp}(G)$ maximizes the performance. In the limit as the number of subordinates $n \rightarrow \infty$, the promoted subordinate is therefore least-biased with probability one in every period. Thus, in contrast to the finite- n dynamics in Theorem 1, bias is eliminated in one step in the large-team limit.

5. The Complements Case

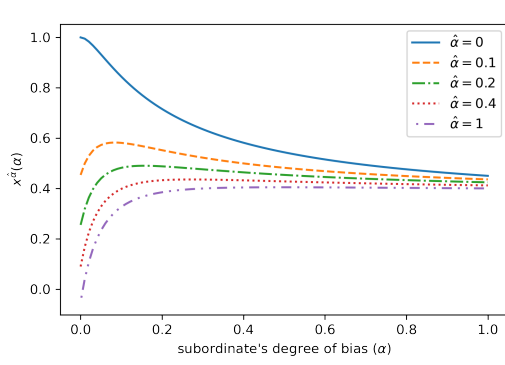
This section focuses on the case where coordination efforts are complements. As in the substitutes case, the first subsection characterizes the relationships among effort allocation, performance, and the degree of bias. The second subsection investigates the dynamics of bias.

5.1. Equilibrium Properties

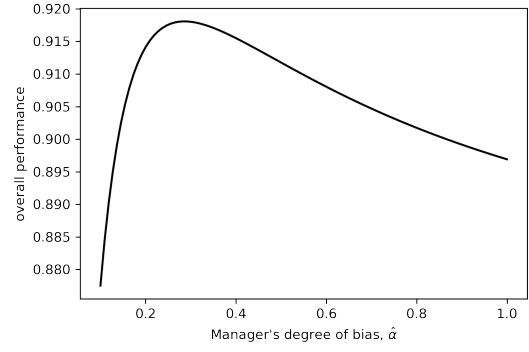
The following lemma summarizes how exerted efforts and performance depend on the degrees of bias of subordinates and managers.

Lemma 3. *Suppose that $v_{12} > 0$. Then, we have the following properties:*

- (a) $\frac{\partial e_s^\alpha}{\partial \alpha} \geq 0$, and $\frac{\partial e_m^{\hat{\alpha}}}{\partial \hat{\alpha}} \geq 0$. The inequality is strict if $e_s^\alpha < 1$ and $e_m^{\hat{\alpha}} < 1$.
- (b) $x^{\hat{\alpha}}(\alpha)$ is non-increasing in $\hat{\alpha}$, and strictly decreasing if $e_m^{\hat{\alpha}} < 1$. Moreover, it is single-peaked with respect to α .
- (c) $\arg \max_\alpha x^{\hat{\alpha}}(\alpha) \leq \hat{\alpha}$. The inequality is strict if $\hat{\alpha} > 0$.



(a) Subordinate's performance ($x^{\hat{\alpha}}(\alpha)$) and bias



(b) The overall performance and degree of the manager's bias (G is a uniform distribution)

Figure 1: Single-peakedness of the subordinate's and overall performance

- (d) $\arg \max_{\alpha} x^{\hat{\alpha}}(\alpha)$ is nondecreasing in $\hat{\alpha}$.
- (e) $\frac{\partial^2 x^{\hat{\alpha}}(\alpha)}{\partial \alpha \partial \hat{\alpha}} \geq 0$. The inequality is strict if $e_s^{\alpha} < 1$ and $e_m^{\hat{\alpha}} < 1$.
- (f) $E[\Pi | \hat{\alpha}]$ is single-peaked with respect to $\hat{\alpha}$. Moreover, $\frac{\partial E[\Pi | \hat{\alpha}]}{\partial \hat{\alpha}}|_{\hat{\alpha}=0} > 0$ if G has a positive measure on $(0, 1]$.
- (g) Suppose that $G_1 \succ_{\text{fsd}} G_2$. Let $\hat{\alpha}_k$ be the maximizer of $E[\Pi | \hat{\alpha}]|_{G=G_k}$ for each $k \in \{1, 2\}$. Then, $\hat{\alpha}_1 > \hat{\alpha}_2$.

We now explain the results in Lemma 3 relative to the corresponding results in Lemma 2. First, the effort devoted to one's own task, e_i^{α} , increases with the degree of bias. This pattern is the same as in the substitutes case.

The first major difference concerns the shape of subordinate performance with respect to the subordinate's degree of bias (Lemma 3 (b)–(d)). In the substitutes case, subordinate performance decreases with the subordinate's own bias. By contrast, in the complements case, this relationship depends on the manager's bias. Lemma 3 (b) shows that, for a given managerial bias, subordinate's performance is single-peaked in the subordinate's bias.²² This single-peakedness indicates that, when efforts are complements, insufficient bias causes subordinates to rely too heavily on the manager, while excessive bias leads them to underinvest in coordination. Lemma 3 (c) shows that the bias of the best performer remains lower than that of the manager, indicating that excessive coordination neglect is still detrimental. Finally, Lemma 3 (d) shows that this maximizer is increasing in the managerial bias.

The second difference appears in the cross-derivative of subordinate performance. Lemma 3 (e) shows that, as the manager's bias increases, the relative performance advantage shifts toward more biased

²²Figure 1 provides numerical illustrations of the results in Lemma 3.

subordinates. This pattern reflects the complementarity of coordination efforts.

Another difference lies in the pattern of overall performance, as shown in Lemma 3 (f). In the substitutes case, higher managerial bias monotonically reduces performance. By contrast, in the complements case, overall performance is single-peaked with respect to the manager's bias. In other words, the peak in overall performance is achieved at a strictly positive level of managerial bias, indicating that some degree of coordination-neglect bias is beneficial. Furthermore, Lemma 3 (g) shows that the optimal managerial bias increases in the subordinates' bias.

This single-peakedness arises from the complementarity of efforts in coordination tasks. When subordinates have intermediate levels of bias, they contribute moderate levels of coordination effort. If the managerial bias is excessively strong, the manager exerts insufficient coordination effort, resulting in a waste of subordinates' contributions. In contrast, if managerial bias is too low, the manager overinvests in coordination, expecting more from subordinates than they actually provide. This misalignment in efforts shifts attention toward independent tasks. Consequently, when managerial bias is either excessively high or low, overall performance declines.

5.2. Dynamics of Bias and Performance

We examine how a promoted subordinate's bias depends on the manager's bias. In the complements case, the dynamics of coordination-neglect bias are monotone with respect to the manager's bias distribution.

Proposition 3. *Suppose that $v_{12} > 0$. Then, for any distribution functions G , $\hat{G}$, and $\bar{G}$, if $\hat{G} >_{\text{fsd}} \bar{G}$, $\mathcal{G}_{\hat{G},G} >_{\text{fsd}} \mathcal{G}_{\bar{G},G}$.*

Proposition 3 is the counterpart to Theorem 1 (a), with the direction of selection reversed: in the complements case, *more* biased managers tend to promote *more* biased subordinates. This result follows from the sign of the cross derivative of $x^{\hat{\alpha}}(\alpha)$, as shown in Lemma 3 (e). When the manager is highly biased, a highly biased subordinate is more likely to achieve high performance, making such subordinates relatively more attractive for promotion.

Proposition 3 does not imply that $\hat{G}_t$ is monotone. To establish the monotonicity of the sequence $(\hat{G}_t)_{t \in \mathbb{N}}$, one must compare $\hat{G}$ and $\mathcal{G}_{\hat{G},G}$. Once $\hat{G}_t >_{\text{fsd}} \hat{G}_{t+1} = \mathcal{G}_{\hat{G}_t,G}$ (resp., $\mathcal{G}_{\hat{G}_t,G} >_{\text{fsd}} \hat{G}_t$), $\hat{G}_{t+1} = \mathcal{G}_{\hat{G}_t,G} >_{\text{fsd}} \hat{G}_{t+2} = \mathcal{G}_{\hat{G}_{t+1},G}$. Therefore, $(\hat{G}_t)_t$ is a decreasing (resp., increasing) sequence in the sense of FOSD.

We derive several sufficient conditions for the dominance relation between $\hat{G}$ and $\mathcal{G}_{\hat{G},G}$. In general,

the sequence $(\hat{G}_t)$ with $\hat{G}_0 = G$ can increase or decrease, depending on G . Theorem 2 characterizes this dominance relation when the manager's bias is fixed. Before we state the result, we introduce notion $\succ_{uv}$, which is referred to as *uniform variability order* (Whitt, 1985). When the two distributions share the same mean, uniform variability ordering implies that having higher variance.²³

Theorem 2. *Suppose that $v_{12} > 0$. There are $\hat{\alpha}_1$ and $\hat{\alpha}_2$ such that $\min(\text{supp}(G)) \leq \hat{\alpha}_1 \leq \hat{\alpha}_2 \leq \max(\text{supp}(G))$ that satisfy the following properties:*

Region (a): For $\hat{\alpha} \in (\min(\text{supp}(G)), \hat{\alpha}_1)$, the promoted distribution is dominated by the original in the FOSD order: $G \succ_{\text{fsd}} \mathcal{G}_{\hat{\alpha}, G}$. Selection unambiguously reduces bias.

Region (b): For $\hat{\alpha} \in (\hat{\alpha}_1, \hat{\alpha}_2)$, the two distributions are not FOSD-comparable, but the promoted distribution exhibits less variability: $G \succ_{uv} \mathcal{G}_{\hat{\alpha}, G}$. Selection compresses the bias distribution without unambiguously shifting its mean.

Region (c): For $\hat{\alpha} \in (\hat{\alpha}_2, \max(\text{supp}(G)))$, the promoted distribution dominates the original: $\mathcal{G}_{\hat{\alpha}, G} \succ_{\text{fsd}} G$. Selection unambiguously amplifies bias.

Furthermore, we have the following properties: (i) Region (a) is non-empty. (ii) If region (c) is non-empty, region (b) is non-empty. (iii) If $x^1(0) > x^1(1)$, region (c) is empty.

Theorem 2 shows that meritocratic selection in the complements case operates through three regimes depending on the manager's bias level: a regime in which selection unambiguously reduces bias (region a), a regime in which the bias distribution is compressed without a clear directional shift (region b), and a regime in which selection unambiguously amplifies bias (region c). The amplification regime is non-empty only if complementarity is strong enough that an unbiased subordinate cannot outperform the most biased one under a maximally biased manager, a condition captured by $x^1(0) < x^1(1)$.

The next result translates this three-regime characterization into conditions on the *initial distribution* of subordinate biases under which the sequence of managerial bias distributions moves monotonically in one direction.²⁴

Proposition 4. (a) *Suppose that $x^1(0) > x^1(1)$, which represents a moderate complementarity regime.*

If the initial distribution places sufficient mass on highly biased subordinates, the next generation of managers is less biased than the current pool. The selection mechanism here parallels the substitutes case: meritocratic promotion gradually reduces coordination neglect.

²³Formally, $G \succ_{uv} \mathcal{G}_{\hat{\alpha}, G}$ if the density ratio $\mathcal{G}'_{\hat{\alpha}, G}/g$ is single-peaked and neither distribution first-order stochastically dominates the other.

²⁴The formal statement and proof are deferred to Appendix B.4.

(b) Suppose that $x^1(0) < x^1(1)$, which represents a strong complementarity regime. If the initial distribution is concentrated at low-bias subordinates with a smaller mass of intermediate types, the next generation of managers is more biased than the current pool. Meritocratic promotion amplifies coordination neglect over time.

The key determinant is the strength of complementarity, captured by the comparison between $x^1(0)$ and $x^1(1)$: this comparison indicates whether the unbiased or the most biased subordinate performs better under the most biased manager ($\hat{\alpha} = 1$), the scenario most favorable to high-bias subordinates. If unbiased types can compete here, they can compete at any managerial bias level; if not, the amplification region is non-empty and selection can propagate bias.

Moderate complementarity: When complementarity is moderate ($x^1(0) > x^1(1)$), an unbiased subordinate outperforms the most biased one even under the most biased manager. Because $\hat{\alpha} = 1$ is the most favorable case for high-bias subordinates, the condition $x^1(0) > x^1(1)$ implies that unbiased subordinates are competitive across *all* managerial bias levels. For the unbiased subordinate to win the promotion contest, however, it must not be crowded out by moderately biased types near the performance peak. Proposition 4 (a) shows that if the initial distribution places sufficient mass on highly biased types and the unbiased type therefore faces sufficiently many low-performing competitors, the next generation of managers is less biased than the current pool in the FOSD order, and the sequence $(\hat{G}_t)$ is monotone decreasing.

By Theorem 2, the converse cannot occur in this regime: under moderate complementarity, the bias-amplification region of Theorem 2 is empty, and the managerial bias distribution cannot shift upward in the FOSD order regardless of the initial distribution. This creates a tension with the static optimum. Lemma 3 (g) shows that overall performance is maximized at a strictly positive managerial bias level, and that this optimal level increases when the subordinate population is more biased. In a workforce with many highly biased subordinates, the statically optimal manager is moderately biased: one who does not overinvest in coordination relative to her subordinates. However, meritocratic promotion systematically selects less biased managers, pushing the organization away from the static optimum.

Strong complementarity: When complementarity is strong ($x^1(0) < x^1(1)$), the picture reverses. Under the most biased manager, the most biased subordinates outperform the unbiased type because their effort allocation aligns with the manager's. In this regime, the bias-amplification region of Theorem 2 is nonempty: for sufficiently biased managers, selection favors more biased subordinates.

Whether the overall sequence $(\hat{G}_t)$ increases depends on the shape of the initial distribution. Proposition 4 (b) shows that if the initial distribution concentrates most of its mass on highly biased types with relatively few intermediate types near the performance peak, the next generation of managers is more biased than the current pool with high probability, and the sequence becomes monotone increasing in the FOSD order.

The intuition is as follows. Under a highly biased manager, the single-peakedness of $x^{\hat{\alpha}}(\alpha)$ implies that the best performer is a subordinate whose bias is slightly below the manager's. For this type to win the promotion contest, the initial distribution must have a thin middle relative to its upper tail, leaving few competitors near the performance peak. When this condition holds, highly biased subordinates consistently win promotion, and the organizational culture drifts toward greater coordination neglect over time.

An increasing sequence cannot arise at all managerial bias levels. When the manager is nearly unbiased ($\hat{\alpha} \approx 0$), complementarity makes the unbiased subordinate the best performer regardless of the distribution, so the sequence cannot increase from an unbiased starting point. For the sequence to be overall increasing, a sufficiently large fraction of managers must themselves be biased, which is why Proposition 4 (b) requires the condition to hold with high probability over draws of the manager from G .

Depending on the distribution of subordinates' biases, the sequence of managerial bias distributions can be either increasing or decreasing. Even a well-designed meritocratic promotion system that rewards coordination may thus inadvertently amplify coordination neglect in the complements case.

5.3. Large-organization Limit

We now consider the large-organization limit, in which the pool of potential successors is sufficiently large that the best-performing subordinate type is almost certainly represented in each cohort. In contrast to the finite- n case, the managerial bias distribution decreases monotonically in the limit.

Theorem 3. (a) In the limit $n \rightarrow \infty$, $\hat{G} \succeq_{\text{fSD}} \mathcal{G}_{\hat{G},G}$ for any G and $\hat{G}$.

(b) Suppose that the support of G is a closed interval. Consider the limit $n \rightarrow \infty$. Then, in the limit distribution of managerial bias as $t \rightarrow \infty$, the support converges to $\{\min \text{supp}(G)\}$.

(c) Under the same assumptions of (b), there exists t^* such that if $t \geq t^*$, the overall performance diminishes over time.

Theorem 3 shows that meritocratic promotion in large organizations drives managerial bias down to the minimum bias in the population, even though overall performance is maximized at a strictly

positive interior bias level. Increasingly “rational” leadership can thus paradoxically reduce long-run organizational performance.

The intuition rests on a key property from Lemma 3 (c): because coordination efforts are complements, a subordinate with slightly lower bias than the manager invests more in coordination and captures benefits that exceed her own expectations, outperforming equally or more biased rivals. Formally, the maximizer of $x^{\hat{\alpha}}(\alpha)$ with respect to α is no greater than $\hat{\alpha}$. In a large subordinate pool, at least one such subordinate is almost certainly present, while those with higher bias face negligible probability of promotion. Managerial bias therefore tends to decrease across generations. When the support of G is continuous, the limiting distribution collapses to a Dirac measure at $\min \text{supp}(G)$, regardless of G .

This progressive reduction in managerial bias does not, however, always improve overall performance. The key is that overall performance is single-peaked in managerial bias: as bias degenerates to its minimum value, performance eventually declines.²⁵

One way to halt this evolution is to randomly select a manager from among the current subordinates, regardless of performance. By breaking the link between performance and promotion, random promotion can outperform performance-based systems in preventing the erosion of overall performance when managers’ bias falls well below that of subordinates. Under random promotion, biased managers persist in the organization, maintaining a diverse range of biases and preventing a decline in overall performance. The following corollary is the formal statement.

Corollary 1. *Suppose that $\text{supp}(G) = [0, \bar{\alpha}]$ for some $\bar{\alpha} > 0$. Let $\Pi^{rdm} := \int E[\Pi \mid \hat{\alpha}]g(\hat{\alpha})d\hat{\alpha}$ be the expected performance under random promotion: managerial bias $\hat{\alpha}$ follows distribution G . Then, $\Pi^{rdm} > \lim_{t \rightarrow \infty} \int E[\Pi \mid \hat{\alpha}]\hat{g}_t(\hat{\alpha})d\hat{\alpha}$*

This provides a rationale for promotion schemes that intentionally maintain diversity in managers’ attitudes toward coordination. For example, these schemes can combine performance-based promotions with random rotation or quota systems.

Taken together, the results of Section 4 and Section 5 reveal a sharp contrast between substitutability and complementarity of coordination efforts. In the substitutes case, coordination neglect is entirely detrimental: meritocratic promotion selects less biased managers and (weakly) improves long-run performance. By contrast, in the complements case, moderate bias can enhance performance; however, as shown in Theorem 3 and Corollary 1, performance-based promotion ultimately reduces managerial bias below its optimal level, leading to lower long-run performance.

²⁵In our performance-based promotion system, we consider overall performance as the performance measurement. To enhance the system’s effectiveness, we could focus on either the value of coordination or the outcomes of independent tasks. However, inefficiencies can still arise from managerial biases, regardless of the chosen performance measurement.

While Theorem 3 and Corollary 1 focus on the large-organization limit, we complement this analysis with numerical computations for finite team sizes, reported in the supplementary material. These simulations show that, for moderate team sizes, performance-based promotion can either increase or decrease the average managerial bias, depending on the initial distribution of subordinates' biases. They also illustrate that changes in the cross-sectional variance of bias are tightly linked to changes in performance: performance improvements tend to be larger when promotion both raises the average bias and compresses its cross-sectional dispersion. The supplementary material provides a theoretical rationale for this pattern using Theorem 2. Moreover, the simulations reveal that the performance erosion predicted by Theorem 3 (c) is not merely an asymptotic phenomenon: at $n = 25$ and over ten promotion rounds, a substantial fraction of initial distributions already lead to negative performance changes.

6. Discussion

This section examines the robustness and broader implications of our main results. We consider two extensions of the baseline model: endogenous total effort and alternative specifications of coordination-neglect bias. We demonstrate that the qualitative conclusions remain consistent in both cases. Next, we discuss the empirical relevance of our analysis, focusing on how the mechanism we identify differs from standard accounts of meritocratic selection failure.

Endogenous Total Effort We assume throughout that each player allocates a fixed amount of effort between independent and coordination tasks. A natural extension is to allow players to choose their total effort level as well.

Endogenizing total effort introduces an asymmetry across types. A subordinate with low bias perceives a higher marginal return to coordination effort and therefore raises total effort, while a high-bias subordinate responds less strongly. Unlike in Lemma 1, where total effort is fixed, this asymmetry interacts with promotion incentives: when subordinates internalize the promotion benefit, the prize raises the marginal return to total effort, and the increase is larger for low-bias subordinates because their marginal product of coordination effort is closer to its true value. The qualitative structure of selection nevertheless survives. The cross-derivative property of $x^{\hat{\alpha}}(\alpha)$ that drives our main results depends only on the sign of v_{12} and on the monotonicity of effort allocations in α , both of which are preserved when total effort is endogenous. In particular, the single-peakedness of overall performance with respect to managerial bias, established in Lemma 3 (f), continues to hold, and Corollary 1 remains valid under this extension.

Alternative Bias Specifications Our analysis treats bias as a fixed individual trait that is unaffected by promotion and independent of strategic sophistication. Two natural relaxations of this assumption maintain the qualitative results.

First, if promotion itself attenuates bias, so that a promoted manager's bias is given by a monotone function $\varphi(\alpha) \leq \alpha$ of her bias as a subordinate, then the selection forces we identify are reinforced and the qualitative results in Theorem 1 and Theorem 3 continue to hold.

Second, consider a *fully rational* subordinate who has $\alpha = 0$ and correctly anticipates the distribution $\hat{G}$ of managerial bias, but not the specific realized $\hat{\alpha}$ of the current manager. She solves $\max_{e_s} u(e_s) + \int v(e_s, e_m^{\hat{\alpha}}) d\hat{G}(\hat{\alpha})$, yielding effort e_s^* that is optimal against $\hat{G}$. Because the ex post optimal effort depends on the realized $\hat{\alpha}$, while e_s^* must be chosen before $\hat{\alpha}$ is observed, e_s^* does not maximize realized performance for any particular realization. She optimizes correctly given her information; still, the gap between ex ante and ex post optimality creates room for certain biased subordinates to outperform her ex post. In the complements case, for each $\hat{\alpha} > 0$ there exists a bias level $\alpha^*(\hat{\alpha}) > 0$ at which a naive subordinate's incorrect belief that the manager shares her bias produces the ex post optimal effort under realization $\hat{\alpha}$. A subordinate with this bias outperforms the fully rational type almost surely. Introducing a small mass of fully rational agents therefore does not alter the bias dynamics or welfare results of Section 5. In the substitutes case, the logic reverses: fully rational agents outperform biased subordinates when managers are highly biased, strengthening selection toward low-bias managers without changing the qualitative implications of Section 4.

Empirical Implications The central empirical focus of our model is comparative: the same performance-based promotion system, even when it explicitly rewards coordination output, can yield qualitatively opposite effects on long-run organizational performance, depending on whether managerial and subordinate coordination efforts are substitutes or complements. In substitutable environments, meritocratic selection reduces managerial bias and improves performance; in complementary environments, the same selection mechanism can drive bias below the performance-maximizing level, thereby decreasing performance, particularly when the subordinate pool is large. The relevant empirical question is whether the long-run performance effects of performance-based promotion vary systematically across environments where coordination efforts are plausibly substitutable or complementary. A related design implication is that promotion rules that preserve diversity in managerial coordination styles may outperform strict performance-based promotion in complementary environments.

At first glance, this prediction resembles existing evidence suggesting that performance-based promo-

tions can fail to select the most effective managers, as indicated in the literature on the Peter Principle (e.g., [Benson, Li, and Shue, 2019](#)). However, the mechanism we identify is distinct. Standard accounts attribute selection failures to a misalignment between the promotion metric and the determinants of managerial effectiveness. In our model, when coordination efforts are complementary, selection failures can persist despite promotion criteria that incorporate coordination output and, in some cases, may even be exacerbated by meritocratic selection. Thus, these two mechanisms suggest different empirical patterns: under a metric-mismatch account, selection failures should diminish when promotion criteria better reflect managerial responsibilities; conversely, under our account, they should decrease in substitutable environments but persist in complementary ones.

Direct identification of the modularity of underlying coordination technology faces well-known challenges (e.g., [Athey and Stern, 1998](#)). Therefore, we view our predictions as comparative patterns rather than a precise identification strategy. Within this framework, the asymmetric direction of selection effects—improving performance in one regime while reducing it in another—stands out as the key implication of our analysis, which is not addressed in existing accounts of meritocratic selection failure.

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Appendix A. Bias Distribution of Promoted Subordinates

In this section, we formally derive $\mathcal{G}$, the distribution function of the subordinate’s bias that yields the highest performance among their colleagues. Given that the manager has bias $\hat{\alpha}$, the performance of the subordinate with bias α is $\theta + x^{\hat{\alpha}}(\alpha)$.

Let g^* be the distribution of α (pdf) that yields performance $\pi_i = \pi'$, which is calculated as follows:

$$g^*(\alpha \mid \hat{\alpha}, \pi') = g(\alpha) \frac{f(\pi' - x^{\hat{\alpha}}(\alpha))}{\int f(\pi' - x^{\hat{\alpha}}(\alpha')) dG(\alpha')}.$$

The term $f(\pi' - x^{\hat{\alpha}}(\alpha))$ is the probability density function that the performance is π' conditioned on the states of the manager and the subordinate. Define $G(\alpha \mid \hat{\alpha}, \pi) = \int^{\alpha} g^*(\alpha' \mid \hat{\alpha}, \pi) d\alpha'$.

Let $H(\pi \mid \hat{\alpha}, G)$ be the probability that the subordinate’s performance is less than π given that the

manager's bias is $\hat{\alpha}$, which is represented by²⁶

$$H(\pi | \hat{\alpha}, G) = \int_{\pi' \in (-\infty, \pi]} \left[\int_{\alpha' \in [0, 1]} f(\pi' - x^{\hat{\alpha}}(\alpha')) dG(\alpha') \right] d\pi'.$$

Then, let $\mathcal{G}_{\hat{\alpha}, G}(\cdot)$ denote the distribution of α of the promoted subordinate who yields the highest performance among n subordinates. Then, $\mathcal{G}_{\hat{\alpha}, G}(\cdot)$ is calculated as follows:

$$\mathcal{G}_{\hat{\alpha}, G}(\alpha) := \int G(\alpha | \hat{\alpha}, \pi) \underbrace{nH'(\pi | \hat{\alpha}, G)[H(\pi | \hat{\alpha}, G)]^{n-1}}_{\text{density that the highest performance is } \pi} d\pi.$$

Finally, $\mathcal{G}_{\hat{G}, G}$ is defined as $\mathcal{G}_{\hat{G}, G}(\alpha) = \int \mathcal{G}_{\hat{\alpha}, G}(\alpha) d\hat{G}(\hat{\alpha})$ for each α .

The distribution has the following properties, which are used in the proofs of some theorems.

Claim 1.

- (i) Suppose that $\frac{\partial^2 x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha} \partial \alpha} \leq 0$ for any $\hat{\alpha}, \alpha$ and strict inequality holds for some α . Then, for any $\hat{\alpha}, \hat{\alpha}'$ with $\hat{\alpha} < \hat{\alpha}'$, $\mathcal{G}_{\hat{\alpha}, G} >_{\text{fsd}} \mathcal{G}_{\hat{\alpha}', G}$.
- (ii) Suppose that $\frac{\partial^2 x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha} \partial \alpha} \geq 0$ for any $\hat{\alpha}, \alpha$ and strict inequality holds for some α . Then, for any $\hat{\alpha}, \hat{\alpha}'$ with $\hat{\alpha} < \hat{\alpha}'$, $\mathcal{G}_{\hat{\alpha}', G} >_{\text{fsd}} \mathcal{G}_{\hat{\alpha}, G}$.

Proof of Claim 1. By differentiating $\mathcal{G}_{\hat{\alpha}, G}(\tilde{\alpha})$ by $\hat{\alpha}$,

$$\begin{aligned} \frac{\partial \mathcal{G}_{\hat{\alpha}, G}(\tilde{\alpha})}{\partial \hat{\alpha}} &= \int_{\alpha \in [0, \tilde{\alpha}]} g(\alpha) \int_{\pi \in \mathbb{R}} \left[-\frac{\partial x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha}} f'(\pi - x^{\hat{\alpha}}(\alpha)) H(\pi | \hat{\alpha}, G) \right. \\ &\quad \left. + (n-1)f(\pi - x^{\hat{\alpha}}(\alpha)) H_{\hat{\alpha}}(\pi | \hat{\alpha}, G) \right] (H(\pi | \hat{\alpha}, G))^{n-2} n d\pi d\alpha \end{aligned}$$

where $H_{\hat{\alpha}}$ is a derivative of H with respect to $\hat{\alpha}$;

$$H_{\hat{\alpha}}(\pi | \hat{\alpha}, G) = - \int_{\alpha \in [0, 1]} \frac{\partial x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha}} \int_{\pi' \in [0, \pi]} f'(\pi' - x^{\hat{\alpha}}(\alpha)) d\pi' d\alpha = - \int_{\alpha \in [0, 1]} \frac{\partial x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha}} f(\pi - x^{\hat{\alpha}}(\alpha)) d\alpha.$$

Note that integration by parts yields

$$\begin{aligned} &\int_{\alpha \in [0, \tilde{\alpha}]} g(\alpha) \int_{\pi \in \mathbb{R}} \frac{\partial x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha}} [f'(\pi - x^{\hat{\alpha}}(\alpha)) H(\pi | \hat{\alpha}, G) \\ &\quad + (n-1)f(\pi - x^{\hat{\alpha}}(\alpha)) H'(\pi | \hat{\alpha}, G)] (H(\pi | \hat{\alpha}, G))^{n-2} n d\pi d\alpha \\ &= \int_{\alpha \in [0, \tilde{\alpha}]} g(\alpha) \left[\frac{\partial x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha}} f(\pi - x^{\hat{\alpha}}(\alpha)) (H(\pi | \hat{\alpha}, G))^{n-1} n \right]_{-\infty}^{\infty} d\alpha = 0. \end{aligned} \tag{4}$$

²⁶To clarify the variable and limits of integration, we indicate the integrating variable and its range in the subscript of the integral sign (e.g., $\int_{\pi' \in \mathbb{R}} f(\pi') d\pi'$), particularly in expressions involving multiple nested integrals or nontrivial bounds. Otherwise, we use the conventional notation.

As (4) is 0, adding (4) to $\frac{\partial \mathcal{G}_{\hat{\alpha}, G}(\tilde{\alpha})}{\partial \hat{\alpha}}$ yields

$$\begin{aligned} \frac{\partial \mathcal{G}_{\hat{\alpha}, G}(\tilde{\alpha})}{\partial \hat{\alpha}} &= \int_{\alpha \in [0, \tilde{\alpha}]} g(\alpha) \int_{\pi \in \mathbb{R}} \left[b_{\pi, \hat{\alpha}}(\alpha) H_{\hat{\alpha}}(\pi | \hat{\alpha}, G) + \frac{\partial x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha}} f(\pi - x^{\hat{\alpha}}(\alpha)) H'(\pi | \hat{\alpha}, G) \right] \\ &\quad \times (H(\pi | \hat{\alpha}, G))^{n-2} n(n-1) d\pi d\alpha. \end{aligned}$$

Note that because $\mathcal{G}_{\hat{\alpha}, G}$ is a cumulative density function on $[0, 1]$, $\mathcal{G}_{\hat{\alpha}, G}(1) = 1$ and $\mathcal{G}_{\hat{\alpha}, G}(0) = 0$, and then

$$\frac{\partial \mathcal{G}_{\hat{\alpha}, G}(1)}{\partial \hat{\alpha}} = \frac{\partial \mathcal{G}_{\hat{\alpha}, G}(0)}{\partial \hat{\alpha}} = 0. \text{ Moreover,}$$

$$\begin{aligned} \frac{\partial \mathcal{G}_{\hat{\alpha}, G}(\tilde{\alpha})}{\partial \hat{\alpha}} &= \int_{\alpha \in [0, \tilde{\alpha}]} g(\alpha) \int_{\alpha' \in [0, 1]} g(\alpha') \int_{\pi \in \mathbb{R}} f(\pi - x^{\hat{\alpha}}(\alpha)) f(\pi - x^{\hat{\alpha}}(\alpha')) \\ &\quad \times \left[\frac{\partial x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha}} - \frac{\partial x^{\hat{\alpha}}(\alpha')}{\partial \hat{\alpha}} \right] (H(\pi | \hat{\alpha}, G))^{n-2} n(n-1) d\pi d\alpha' d\alpha \\ &= \begin{cases} \int_{\pi \in \mathbb{R}} W(1 | \pi) \times W(\tilde{\alpha} | \pi) \times (H(\pi | \hat{\alpha}, G))^{n-2} n(n-1) \\ \quad \times \left[\int_{\alpha \in [0, 1]} w(\alpha | \pi, \tilde{\alpha}) \frac{\partial x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha}} d\alpha - \int_{\alpha' \in [0, 1]} w(\alpha' | \pi) \frac{\partial x^{\hat{\alpha}}(\alpha')}{\partial \hat{\alpha}} d\alpha' \right] d\pi \end{cases} \end{aligned} \quad (5)$$

where

$$\begin{aligned} w(\hat{\alpha} | \pi) &= \frac{g(\alpha') f(\pi - x^{\hat{\alpha}}(\alpha'))}{W(1 | \pi)}, \quad W(\tilde{\alpha} | \pi) = \int_0^{\tilde{\alpha}} g(\alpha') f(\pi - x^{\hat{\alpha}}(\alpha')) d\alpha', \\ \text{and } w(\alpha' | \pi, \tilde{\alpha}) &= \begin{cases} \frac{g(\alpha') f(\pi - x^{\hat{\alpha}}(\alpha'))}{W(\tilde{\alpha} | \pi)} & \text{if } \alpha' < \tilde{\alpha}, \\ 0 & \text{if } \alpha' \geq \tilde{\alpha}. \end{cases} \end{aligned}$$

Note that both $w(\hat{\alpha} | \pi)$ and $w(\hat{\alpha} | \pi, \tilde{\alpha})$ are probability density functions with respect to α' and $w(\hat{\alpha} | \pi)$ first-order stochastically dominates $w(\hat{\alpha} | \pi, \tilde{\alpha})$ as they are single-crossing (see, e.g., [Shaked and Shanthikumar, 2007](#), Theorem 1.A.12).

Now, suppose that $\frac{\partial^2 x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha} \partial \alpha} \leq 0$ and strict inequality holds for some α . Then, since $w(\hat{\alpha} | \pi)$ first-order stochastically dominates $w(\hat{\alpha} | \pi, \tilde{\alpha})$, the second term of (5) is positive for each π . Therefore, $\mathcal{G}_{\hat{\alpha}', G}(\tilde{\alpha}) < \mathcal{G}_{\hat{\alpha}, G}(\tilde{\alpha})$ for each $\tilde{\alpha}$, $\hat{\alpha}$, and $\hat{\alpha}'$ with $\hat{\alpha} > \hat{\alpha}'$. This implies the first-order stochastic dominance with a lower $\hat{\alpha}$. If $\frac{\partial^2 x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha} \partial \alpha} \geq 0$, the reverse order holds. $\square$

Appendix B. Proofs

Appendix B.1. Proofs in Section 3

Proof of Proposition 2. Note that the solution is positive by Assumption 1 (iii); $e_m = 0$ and $e_s = 0$ are not an optimal solution. Especially, if $\beta = 0$, $e_m = 1$ and $e_s = 1$ are also not an optimal solution.

Consider a player whose bias degree is $\beta \in [0, 1]$. Then, equation (2) determines each player's optimal effort level.

By the assumption of Proposition 2 and concavity of u and v , for each e_m , there uniquely exists $\varphi(e_m) \in [0, 1]$ such that $(\varphi(e_m) - 1)[(1 - \beta)v_1(\varphi(e_m), e_m) + u'(\varphi(e_m))] = 0$, which allows a corner solution $\varphi(e_m) = 1$. Similarly, for each e_s , there also uniquely exists $\psi(e_s) \in [0, 1]$ such that $(\psi(e_s) - 1)[u'(\psi(e_s)) + (1 - \beta)v_2(e_s, \psi(e_s))] = 0$. The solution to (2) satisfies $\varphi(e_m) = e_s$ and $\psi(e_s) = e_m$. By $v_{11} \leq 0, v_{22} \leq 0$ and $u'' \leq 0$, the implicit function theorem implies that ψ and φ are continuous, which implies the existence of the solution.

The uniqueness follows from the concavity assumption.²⁷ We have three cases.

Case (i): If (2) has multiple interior solutions, $(e_s, e_m), (e'_s, e'_m)$, the Taylor expansion implies that for some $(\tilde{e}_s, \tilde{e}_m)$,

$$D \begin{bmatrix} e_s - e'_s \\ e_m - e'_m \end{bmatrix} = 0, \quad \text{where } D = \begin{bmatrix} (1 - \beta)v_{11}(\tilde{e}_s, \tilde{e}_m) + u''(\tilde{e}_s) & (1 - \beta)v_{12}(\tilde{e}_s, \tilde{e}_m) \\ (1 - \beta)v_{21}(\tilde{e}_s, \tilde{e}_m) & u''(\tilde{e}_m) + (1 - \beta)v_{22}(\tilde{e}_s, \tilde{e}_m) \end{bmatrix}$$

Assumption 1 implies that $\det(D) > 0$, which implies that $e_s = e'_s$ and $e_m = e'_m$.

Case (ii): Suppose that (e_s, e_m) is a corner solution with $e_s = 1$ and $e_m < 1$. Then, $u'(1) + (1 - \beta)v_1(1, e_m) = \varepsilon$ and $u'(e_m) + (1 - \beta)v_2(1, e_m) = 0$ for some $\varepsilon > 0$.

If there is an interior solution (e'_s, e'_m) that satisfying $D(1 - e'_s, e_m - e'_m)^\top = (\varepsilon, 0)^\top$. Then, the solution to the equation satisfies $1 - e'_s \leq 0$, which is a contradiction.

Next, if there is another corner solution $(e'_s, 1)$, $D(1 - e'_s, e_m - 1)^\top = (\varepsilon, -\varepsilon')^\top$. Then, if $v_{12} < 0$, the solution to the above equation implies that $1 - e_s < 0$, which is a contradiction. If $v_{12} > 0$,

$$\varepsilon = u'(1) + (1 - \alpha)v_1(1, e_m) \leq u'(e'_s) + (1 - \alpha)v_1(e'_s, e_m) \leq u'(e'_s) + (1 - \alpha)v_1(e'_s, 1) = 0,$$

which is a contradiction.

The case in which (e_s, e_m) is a corner solution with $e_s < 1$ and $e_m = 1$ is proved in the same manner.

Case (iii): Suppose that (e_s, e_m) is a corner solution with $e_s = 1$ and $e_m = 1$.

If another solution exists, it is an interior solution as the best response is unique. Let (e'_s, e'_m) be the interior solution. Then, $D(1 - e'_s, 1 - e'_m)^\top = (\varepsilon, \varepsilon')^\top$, for some $\varepsilon, \varepsilon' > 0$. If $v_{12} > 0$, the above equation

²⁷One could also appeal to the Gale-Nikaidô univalence theorem to establish uniqueness. We instead present a direct argument that avoids dealing with the Jacobian at boundary points and may be more transparent for readers unfamiliar with univalence theorems.

implies that $1 - e_s < 0$.

If $v_{12} < 0$, $\varepsilon = u'(1) + (1 - \beta)v_1(1, 1) \leq u'(e'_s) + (1 - \beta)v_1(e'_s, 1) \leq u'(e'_s) + (1 - \beta)v_1(e'_s, e'_m) = 0$, which is a contradiction. $\square$

Appendix B.2. Proofs in Section 4

Proof of Lemma 2. Proof of (a): By a calculation, if (e_s^α, e_m^α) is an interior solution,

$$\begin{aligned} \frac{\partial e_s^\alpha}{\partial \alpha} &= \frac{v_1(u''(e_m) + (1 - \alpha)v_{22}) - (1 - \alpha)v_2v_{12}}{\det(D)}, \\ \frac{\partial e_m^\alpha}{\partial \alpha} &= \frac{v_2((1 - \alpha)v_{11} + u''(e_s)) - (1 - \alpha)v_{21}v_1}{\det(D)}, \end{aligned} \quad (6)$$

where $D = \begin{pmatrix} (1 - \alpha)v_{11} + u''(e_s) & (1 - \alpha)v_{12} \\ (1 - \alpha)v_{21} & u''(e_m) + (1 - \alpha)v_{22} \end{pmatrix}$.

By Assumption 1 (ii), $\det(D) > 0$. Also by Assumption 1 (iv), $u'' < \max\{v_2/v_1, v_1/v_2\}v_{12}$. Then, as $v_i < 0$, $v_{ii} < 0$ and $v_{12} < 0$,

$$v_1(u''(e_m) + (1 - \alpha)v_{22}) - (1 - \alpha)v_2v_{12} > v_1u''(e_m) - (1 - \alpha)v_2v_{12} > v_2v_{12} - (1 - \alpha)v_2v_{12} > 0.$$

This shows that $\frac{\partial e_s^\alpha}{\partial \alpha} > 0$. A similar calculation shows that $\frac{\partial e_m^\alpha}{\partial \alpha} > 0$.

Consider the case where (e_s^α, e_m^α) is a corner solution with $e_s^\alpha = 1$ and $e_m^\alpha < 1$. Then, e_m^α is the solution to $(1 - \alpha)v_2(1, e_m) + u'(e_m) = 0$, which implies that $\frac{\partial e_m^\alpha}{\partial \alpha} = v_2/(v_{22} + u''(e_m^\alpha)) > 0$.

We now check $e_s^{\alpha'} = 1$ is optimal for $\alpha' > \alpha$. Note that $u'(1) + (1 - \alpha)v_1(1, e_m^\alpha) \geq 0$. Further, by $v_{12} < 0$ and $u'' < (v_2/v_1)v_{12}$, $\frac{d}{d\alpha}[u'(1) + (1 - \alpha)v_1(1, e_m^\alpha)] = -v_1 + v_{12}\frac{v_2}{v_{22} + u''(e_m^\alpha)} > 0$. Therefore, $u'(1) + (1 - \alpha')v_1(1, e_m^{\alpha'}) > 0$ for each $\alpha' > \alpha$. In the case where (e_s^α, e_m^α) is a corner solution with $e_s^\alpha < 1$ and $e_m^\alpha = 1$, the monotonicity is proved in the same manner.

Consider the case where $(e_s^\alpha, e_m^\alpha) = (1, 1)$ is a corner solution. Then, $u'(1) + (1 - \alpha)v_1(1, 1) \geq 0$ and $u'(1) + (1 - \alpha)v_2(1, 1) \geq 0$. As $v_1, v_2 \leq 0$, these inequalities hold for higher α . Therefore, $(e_s^{\alpha'}, e_m^{\alpha'}) = (1, 1)$.

Proof of (b): Note that $u(e) + v(e, e_m)$ is concave in e , and decreasing in e_m . By Lemma 2 (a), e_m^α and e_s^α are increasing in α , and $x^{\hat{\alpha}}(\alpha)$ is decreasing in $\hat{\alpha}$.

We now show that $x^{\hat{\alpha}}(\alpha)$ is decreasing in α . The derivative of $x^{\hat{\alpha}}(\alpha)$ with respect to α is calculated as $\frac{\partial x^{\hat{\alpha}}(\alpha)}{\partial \alpha} = \frac{\partial e_s^\alpha}{\partial \alpha} [u'(e_s^\alpha) + v_1(e_s^\alpha, e_m^{\hat{\alpha}})]$. Consider $\alpha = 0$. Then, by the first-order condition and $v_{12} < 0$,

$u'(e_s^0) = -v_1(e_s^0, e_s^0) < -v_1(e_s^0, e_s^{\hat{\alpha}})$, which implies that $u'(e_s^0) + v_1(e_s^0, e_s^{\hat{\alpha}}) < 0$. By concavity and e_s^α is increasing in α , $u'(e_s^\alpha) + v_1(e_s^\alpha, e_s^{\hat{\alpha}}) < 0$. Therefore, $\frac{\partial x^{\hat{\alpha}}(\alpha)}{\partial \alpha} < 0$ for any $\alpha \in [0, 1]$.

Proof of (c): As for the cross derivative, $\frac{\partial^2 x^{\hat{\alpha}}(\alpha)}{\partial \alpha \partial \hat{\alpha}} = \frac{\partial e_s^\alpha}{\partial \alpha} \frac{\partial e_m^{\hat{\alpha}}}{\partial \hat{\alpha}} v_{12}(e_s^\alpha, e_m^{\hat{\alpha}}) < 0$.

Proof of (d): First note that $E[\Pi | \hat{\alpha}]$ is concave with respect to $e_m^{\hat{\alpha}}$. As $e_m^{\hat{\alpha}}$ is increasing in $\hat{\alpha}$, $E[\Pi | \hat{\alpha}]$ is single-peaked with respect to $\hat{\alpha}$. Moreover, $\frac{\partial E[\Pi | \hat{\alpha}]}{\partial \hat{\alpha}} < 0$ at $\hat{\alpha} = 0$. To show this, note that

$$E[\Pi | \hat{\alpha}] = u(e_m^{\hat{\alpha}}) + \int [u(e_s^\alpha) + v(e_s^\alpha, e_m^{\hat{\alpha}})] d\alpha, \quad \frac{\partial E[\Pi | \hat{\alpha}]}{\partial \hat{\alpha}} = \frac{\partial e_m^{\hat{\alpha}}}{\partial \hat{\alpha}} \left[u'(e_m^{\hat{\alpha}}) + \int v_2(e_s^\alpha, e_m^{\hat{\alpha}}) d\alpha \right].$$

Note also that $u(e_m^0) + v_2(e_s^0, e_m^0) = 0$ by the first-order condition for the manager's problem. Then, by $v_{12} < 0$, and e_s^α is increasing in α , for each $\alpha > 0$, $0 = u(e_m^0) + v_2(e_s^0, e_m^0) > u(e_m^0) + v_2(e_s^\alpha, e_m^0)$. This shows that $\frac{\partial E[\Pi | \hat{\alpha}]}{\partial \hat{\alpha}} < 0$. $\square$

Proof of Theorem 1. Proof of (a): By Lemma 2, $\frac{\partial^2 x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha} \partial \alpha} < 0$. Then, Claim 1 (i), $\mathcal{G}_{\hat{\alpha}, G}(\tilde{\alpha})$ is increasing in $\hat{\alpha}$. Therefore, if $\hat{G} \succeq_{\text{fsd}} \tilde{G}$, $\mathcal{G}_{\hat{G}, G}(\tilde{\alpha}) = \int \mathcal{G}_{\hat{\alpha}, G}(\tilde{\alpha}) d\hat{G}(\hat{\alpha}) \geq \mathcal{G}_{\tilde{G}, G}(\tilde{\alpha})$, which completes the proof.

Proof of (b): Before the proof, we provide the following claim:

Claim 2. Define $\sigma_{\hat{\alpha}}(\alpha)$ for each $\hat{\alpha}, \alpha \in [0, 1]$ as follows.

$$\sigma_{\hat{\alpha}}(\alpha) = \int_{\pi \in \mathbb{R}} f(\pi) n \left[\int_{\alpha' \in [0, 1]} F(\pi + x^{\hat{\alpha}}(\alpha) - x^{\hat{\alpha}}(\alpha')) dG(\alpha') \right]^{n-1} d\pi.$$

If $v_{12} < 0$, the following properties hold: (i) $\mathcal{G}'_{\hat{\alpha}, G}(\alpha) > g(\alpha)$ if and only if $\sigma_{\hat{\alpha}}(\alpha) > 1$. (ii) $\sigma_{\hat{\alpha}}(\alpha)$ is decreasing in α .

Proof of Claim 2. Proof of (i): Note that the density of $\mathcal{G}_{\hat{\alpha}, G}$ is rewritten as

$$\begin{aligned} \mathcal{G}'_{\hat{\alpha}, G}(\alpha) &= g(\alpha) \left[\int_{\pi \in \mathbb{R}} f(\pi - x^{\hat{\alpha}}(\alpha)) n \left[\int_{\alpha' \in [0, 1]} F(\pi - x^{\hat{\alpha}}(\alpha')) dG(\alpha') \right]^{n-1} d\pi \right] \\ &= g(\alpha) \left[\int_{\pi \in \mathbb{R}} f(\pi) n \left[\int_{\alpha' \in [0, 1]} F(\pi + x^{\hat{\alpha}}(\alpha) - x^{\hat{\alpha}}(\alpha')) dG(\alpha') \right]^{n-1} d\pi \right]. \end{aligned}$$

Then, $\mathcal{G}'_{\hat{\alpha}, G}(\alpha) > g(\alpha)$ if and only if $\sigma_{\hat{\alpha}}(\alpha) > 1$.

Proof of (ii): Note that since $x^{\hat{\alpha}}(\alpha)$ is decreasing in α (Lemma 2 (b)) and F is increasing, $\sigma_{\hat{\alpha}}(\alpha)$ is decreasing in α . $\square$

By Claim 2 (ii), $\sigma_{\hat{\alpha}}(\alpha)$ is decreasing in α . Furthermore, as $\int \sigma_{\hat{\alpha}}(\alpha) dG(\alpha) = \int \mathcal{G}'_{\hat{\alpha}, G}(\alpha) d\alpha = 1$, there is $\bar{\alpha}$ such that $\sigma_{\hat{\alpha}}(\alpha) > 1$ if and only if $\alpha < \bar{\alpha}$. Equivalently, $\mathcal{G}'_{\hat{\alpha}, G}(\alpha) > g(\alpha)$ if and only if $\alpha < \bar{\alpha}$. By

Theorem 1.A.12 of [Shaked and Shanthikumar \(2007\)](#), this implies that $G \succ_{\text{fsd}} \mathcal{G}_{\hat{\alpha}, G}$. As this relationship holds for any $\hat{\alpha}$, $G \succ_{\text{fsd}} \mathcal{G}_{G', G}$ for any distribution of $\hat{\alpha}$, G' .

Proof of (c): Note that the following formula gives the evolution dynamics: $\hat{G}_{t+1} = \mathcal{G}_{\hat{G}_t, G}$. By Theorem 1 (b), we have $G \succ_{\text{fsd}} \hat{G}_1$. Then, by Theorem 1 (a), $\hat{G}_2 = \mathcal{G}_{\hat{G}_1, G} \succ_{\text{fsd}} \mathcal{G}_{G, G} = \hat{G}_1$. Again by Theorem 1 (b), we have $G \succ_{\text{fsd}} \hat{G}_2$. Similarly, since $\hat{G}_3 = \mathcal{G}_{\hat{G}_2, G}$ and $G \succ_{\text{fsd}} \hat{G}_2 \succ_{\text{fsd}} \hat{G}_1$, $G \succ_{\text{fsd}} \hat{G}_2 \succ_{\text{fsd}} \hat{G}_3 \succ_{\text{fsd}} \hat{G}_1$.

To complete the proof, we proceed by induction. Assume the induction hypothesis: (i) If t is even, $\hat{G}_{t-2} \succ_{\text{fsd}} \hat{G}_t \succ_{\text{fsd}} G_{t-1}$, (ii) If t is odd, $\hat{G}_{t-1} \succ_{\text{fsd}} \hat{G}_t \succ_{\text{fsd}} \hat{G}_{t-2}$.

Consider $t+1$. Then, if t is even, $t+1$ is odd. Then, $\hat{G}_{t+1} = \mathcal{G}_{\hat{G}_t, G}$, and $\hat{G}_t = \mathcal{G}_{\hat{G}_{t-1}, G}$. As $\hat{G}_t \succ_{\text{fsd}} \hat{G}_{t-1}$, it follows that $\hat{G}_t \succ_{\text{fsd}} G_{t+1}$. Also, as $\hat{G}_{t-1} = \mathcal{G}_{\hat{G}_{t-2}, G}$ and $\hat{G}_{t-2} \succ_{\text{fsd}} \hat{G}_t$, $\hat{G}_{t+1} \succ_{\text{fsd}} \hat{G}_t$. This confirms the induction step of (ii). A similar argument applies for (i) when t is odd.

Proof of (d): We consider $\hat{G}_t = \mathcal{G}_{\hat{G}_{t-1}, G}$ as a Markov chain whose state variable is manager's bias, $\hat{\alpha}_t$. Then, as $\mathcal{G}'_{\hat{\alpha}, G}(\alpha) > 0$ for any $\alpha, \hat{\alpha} \in [0, 1]$, the probability measure corresponding $\mathcal{G}$ satisfies the Doeblin's condition. This full-supportness ensures that the Markov chain is aperiodic and irreducible. Therefore, $\hat{G}_t$ has a convergent.²⁸ $\square$

Appendix B.3. Proofs in Section 5

Proof of Lemma 3. **Proof of (a):** As in the proof of Lemma 2 (a), the interior solution is characterized as (6). Then, as $v_{12} > 0$, $v_i v_{ii} < 0$, and $u'' < 0$, $\frac{\partial e_s^\alpha}{\partial \alpha} > 0$ and $\frac{\partial e_m^\alpha}{\partial \alpha} > 0$.

Consider the case where (e_s^α, e_m^α) is a corner solution with $e_s^\alpha = 1$ and $e_m^\alpha < 1$. Then, e_m^α is the solution to $(1 - \alpha)v_2(1, e_m) + u'(e_m) = 0$, which implies that $\frac{\partial e_m^\alpha}{\partial \alpha} = v_2/(v_{22} + u''(e_m)) > 0$.

We now check $e_s^{\alpha'} = 1$ is optimal for $\alpha' > \alpha$. Note that $u'(1) + (1 - \alpha)v_1(1, e_m^\alpha) \geq 0$. Further, by $v_{12} > 0$ and $u'' < 0$, $\frac{d}{d\alpha}[u'(1) + (1 - \alpha)v_1(1, e_m^\alpha)] > 0$. Therefore, $u'(1) + (1 - \alpha')v_1(1, e_m^{\alpha'}) > 0$ for each $\alpha' > \alpha$. In the case where (e_s^α, e_m^α) is a corner solution with $e_s^\alpha < 1$ and $e_m^\alpha = 1$, the monotonicity is proved in the same manner.

Consider the case where $(e_s^\alpha, e_m^\alpha) = (1, 1)$ is a corner solution. Then, $u'(1) + (1 - \alpha)v_1(1, 1) \geq 0$ and $u'(1) + (1 - \alpha)v_2(1, 1) \geq 0$. As $v_1, v_2 \leq 0$, these inequalities hold for higher α . Therefore, $(e_s^{\alpha'}, e_m^{\alpha'}) = (1, 1)$.

Proof of (b): Note that $u(e) + v(e, e_m)$ is strictly concave with respect to e and is decreasing in e_m . By Lemma 3 (a), $e_m^{\hat{\alpha}}$ and e_s^α are nondecreasing $\hat{\alpha}$ and α , respectively, and strictly increasing if $e_m^{\hat{\alpha}} < 1$ and

²⁸See, for example, Theorem 16.2.3 of [Meyn and Tweedie \(1993\)](#).

$e_s^\alpha < 1$. Consequently, $x^{\hat{\alpha}}(\alpha)$ is non-increasing in $\hat{\alpha}$ and single-peaked with respect to α .²⁹

Proof of (c): Consider the maximizer, $\arg \max_\alpha x^{\hat{\alpha}}(\alpha)$. By the first-order condition, if $e_s^{\hat{\alpha}} < 1$, $u'(e_s^{\hat{\alpha}}) = -(1 - \hat{\alpha})v_1(e_s^{\hat{\alpha}}, e_m^{\hat{\alpha}})$. If $e_s^{\hat{\alpha}} = 1$, by the monotonicity of e_s^α , there is $\alpha^* \leq \hat{\alpha}$ such that $u'(e_s^{\alpha^*}) = -(1 - \alpha^*)v_1(e_s^{\alpha^*}, e_m^{\alpha^*})$, and for each $\alpha \geq \alpha^*$, $e_s^\alpha = 1$.

Therefore, we can focus on α^* , and

$$\frac{\partial x^{\hat{\alpha}}(\alpha^*)}{\partial \alpha} = \frac{\partial e_s^{\alpha^*}}{\partial \alpha} \left[[v_1(e_s^{\alpha^*}, e_m^{\hat{\alpha}}) - v_1(e_s^{\alpha^*}, e_m^{\alpha^*})] + \alpha v_1(e_s^{\alpha^*}, e_m^{\alpha^*}) \right].$$

If $\alpha^* = \hat{\alpha}$, $\frac{\partial x^{\hat{\alpha}}(\alpha^*)}{\partial \alpha} < 0$. If $\alpha^* < \hat{\alpha}$, as $e_s^{\hat{\alpha}} = e_s^{\alpha^*} = 1$, $v_1(e_s^{\alpha^*}, e_m^{\hat{\alpha}}) = v_1(e_s^{\alpha^*}, e_m^{\alpha^*})$. Then, $\frac{\partial x^{\hat{\alpha}}(\alpha^*)}{\partial \alpha} < 0$. As $x^{\hat{\alpha}}$ is single-peaked, $x^{\hat{\alpha}}(\alpha)$ has a decreasing region in $(\hat{\alpha}, 1)$, and thus the peak is strictly less than $\hat{\alpha}$ unless $\hat{\alpha} = \min \text{supp}(G)$.

Proof of (d): For each $\hat{\alpha}_1, \hat{\alpha}_2$ with $\hat{\alpha}_1 > \hat{\alpha}_2$, let $\alpha_k^* = \arg \max_\alpha x^{\hat{\alpha}_k}(\alpha)$ for each $k \in \{1, 2\}$. Then, at $\alpha = \alpha_k^*$,

$$0 = \frac{\partial x^{\hat{\alpha}_1}(\alpha_1^*)}{\partial \alpha} = \left[u'(e_s^{\alpha_1^*}) + v_1(e_s^{\alpha_1^*}, e_m^{\hat{\alpha}_1}) \right] \frac{\partial e_s^{\alpha_1^*}}{\partial \alpha} \geq \left[u'(e_s^{\alpha_1^*}) + v_1(e_s^{\alpha_1^*}, e_m^{\hat{\alpha}_2}) \right] \frac{\partial e_s^{\alpha_1^*}}{\partial \alpha}$$

by $v_{12} > 0$ and $\frac{\partial e_m^{\hat{\alpha}}}{\partial \alpha} \geq 0$. This implies that $\frac{\partial x^{\hat{\alpha}_2}(\alpha_1^*)}{\partial \alpha} < 0$, and therefore, as $x^{\hat{\alpha}}$ is single-peaked, $\alpha_1^* > \alpha_2^*$. This implies that the maximizer of $x^{\hat{\alpha}}(\cdot)$ is nondecreasing in $\hat{\alpha}$ (strict if $\frac{\partial e_m^{\hat{\alpha}}}{\partial \alpha} > 0$).

Proof of (e): As for the cross derivative, $\frac{\partial^2 x^{\hat{\alpha}}(\alpha)}{\partial \alpha \partial \hat{\alpha}} = \frac{\partial e_s^\alpha}{\partial \alpha} \frac{\partial e_m^{\hat{\alpha}}}{\partial \hat{\alpha}} v_{12} \geq 0$. The inequality is strict if $e_s^{\hat{\alpha}} < 1$ and $e_m^{\hat{\alpha}} < 1$.

Proof of (f): Note that $E[\Pi | \hat{\alpha}]$ is concave with respect to $e_m^{\hat{\alpha}}$. Since $e_m^{\hat{\alpha}}$ is increasing in $\hat{\alpha}$, $E[\Pi | \hat{\alpha}]$ is single-peaked with respect to $\hat{\alpha}$.

To prove that $\frac{\partial E[\Pi | \hat{\alpha}]}{\partial \hat{\alpha}}|_{\hat{\alpha}=0} > 0$, note that by the objective function of the manager with $\hat{\alpha} = 0$, $u'(e_m^0) + v_2(e_s^0, e_m^0) = 0$. Using this equation,

$$\frac{\partial E[\Pi | \hat{\alpha}]}{\partial \hat{\alpha}} \Big|_{\hat{\alpha}=0} = \left[\int \left[u'(e_m^0) + v_2(e_s^\alpha, e_m^0) \right] dG(\alpha) \right] \frac{\partial e_m^{\hat{\alpha}}}{\partial \hat{\alpha}} \Big|_{\hat{\alpha}=0} > \left[\int \left[u'(e_m^0) + v_2(e_s^0, e_m^0) \right] dG(\alpha) \right] \frac{\partial e_m^{\hat{\alpha}}}{\partial \hat{\alpha}} \Big|_{\hat{\alpha}=0} = 0.$$

Therefore, $\frac{\partial E[\Pi | \hat{\alpha}]}{\partial \hat{\alpha}}|_{\hat{\alpha}=0} > 0$.

Proof of (g): Suppose that $G_1 \succ_{\text{fsd}} G_2$. Recall that $\hat{\alpha}_k$ is the maximizer of $E[\Pi | \hat{\alpha}]|_{G=G_k}$ for each

²⁹Note that the concave function is single-peaked, and any monotonic transformation of a single-peaked function is single-peaked.

$k \in \{1, 2\}$. Then, for any $\hat{\alpha}$, by $v_{21} > 0$.

$$\begin{aligned} \frac{\partial E[\Pi | \hat{\alpha}]}{\partial \hat{\alpha}} \Big|_{G=G_1} &= \left[\int \left[u'(e_m^{\hat{\alpha}}) + v_2(e_s^\alpha, e_m^{\hat{\alpha}}) \right] dG_1(\alpha) \right] \frac{\partial e_m^{\hat{\alpha}}}{\partial \hat{\alpha}} \\ &> \left[\int \left[u'(e_m^{\hat{\alpha}}) + v_2(e_s^\alpha, e_m^{\hat{\alpha}}) \right] dG_2(\alpha) \right] \frac{\partial e_m^{\hat{\alpha}}}{\partial \hat{\alpha}} = \frac{\partial E[\Pi | \hat{\alpha}]}{\partial \hat{\alpha}} \Big|_{G=G_2} \end{aligned}$$

Therefore, as $\frac{\partial E[\Pi | \hat{\alpha}]}{\partial \hat{\alpha}} \Big|_{G=G_1} = 0$ for $\hat{\alpha} = \hat{\alpha}_1$, with the fact that $E[\Pi | \hat{\alpha}]$ is single-peaked, $\hat{\alpha}_1 > \hat{\alpha}_2$. $\square$

Proof of Proposition 3. By Lemma 3 (e), $\frac{\partial^2 x^{\hat{\alpha}}(\alpha)}{\partial \hat{\alpha} \partial \alpha} \geq 0$ and strict inequality holds for some α . Then, the statement follows from Claim 1 (ii). $\square$

Proof of Theorem 2. We assume that $\min(\text{supp}(G)) = 0$ and $\max(\text{supp}(G)) = 1$ for notational simplicity, but this is without loss of generality.

Before the proof, we provide the following claim:

Claim 3. Define $\sigma_{\hat{\alpha}}(\alpha)$ for each $\hat{\alpha}, \alpha \in [0, 1]$ as follows.

$$\sigma_{\hat{\alpha}}(\alpha) = \int_{\pi \in \mathbb{R}} f(\pi) n \left[\int_{\alpha' \in [0, 1]} F(\pi + x^{\hat{\alpha}}(\alpha) - x^{\hat{\alpha}}(\alpha')) dG(\alpha') \right]^{n-1} d\pi.$$

We have the following properties for each $\hat{\alpha}$ and G . (i) $\mathcal{G}'_{\hat{\alpha}, G}(\alpha) > g(\alpha)$ if and only if $\sigma_{\hat{\alpha}}(\alpha) > 1$. (ii) $\sigma_{\hat{\alpha}}(\alpha)$ is single-peaked with respect to α . (iii) If $\max\{\sigma_{\hat{\alpha}}(1), \sigma_{\hat{\alpha}}(0)\} > 1$, then $1 > \min\{\sigma_{\hat{\alpha}}(1), \sigma_{\hat{\alpha}}(0)\}$. (iv) Let $\alpha^* = \arg \max_{\alpha} x^{\hat{\alpha}}(\alpha)$. Then, $\sigma_{\hat{\alpha}}(\alpha^*) > 1$. (v) $\sigma_0(0) > 1$. (vi) $\sigma_{\hat{\alpha}}(0)$ is decreasing in $\hat{\alpha}$ and $\sigma_{\hat{\alpha}}(1)$ is increasing in $\hat{\alpha}$, and; (vii) If $\sigma_1(0) > 1$, $G \succ_{\text{fsd}} \mathcal{G}_{\hat{\alpha}, G}$.

Proof of Claim 3. Proof of (i): The proof is the same as Claim 2 (i).

Proof of (ii): Note that as F is increasing, $\sigma_{\hat{\alpha}}(\alpha)$ is increasing in α if and only if $x^{\hat{\alpha}}(\alpha)$ is increasing in α . Since $x^{\hat{\alpha}}(\alpha)$ is single-peaked with respect to α , $\sigma_{\hat{\alpha}}(\alpha)$ is also single-peaked with respect to α .

Proof of (iii): By Claim 3 (ii), $\sigma_{\hat{\alpha}}(\cdot)$ is single-peaked for each $\hat{\alpha}$. Then, if $\min\{\sigma_{\hat{\alpha}}(1), \sigma_{\hat{\alpha}}(0)\} \geq 1$, $\sigma_{\hat{\alpha}}(\alpha) \geq 1$ for each α .

On the other hand, $\int \sigma_{\hat{\alpha}}(\alpha) g(\alpha) d\alpha = \int \mathcal{G}'_{\hat{\alpha}, G}(\alpha) d\alpha = 1$. Then, $\sigma_{\hat{\alpha}}(\alpha) = 1$ for each α . This is a contradiction.

Proof of (iv): As α^* is the strict maximizer of $x^{\hat{\alpha}}$, $F(\pi + x^{\hat{\alpha}}(\alpha^*) - x^{\hat{\alpha}}(\alpha')) > F(\pi)$ for each $\alpha' \neq \alpha^*$. Then,

$$\sigma_{\hat{\alpha}}(\alpha^*) = \int_{\pi \in \mathbb{R}} f(\pi) n \left[\int_{\alpha' \in [0, 1]} F(\pi + x^{\hat{\alpha}}(\alpha^*) - x^{\hat{\alpha}}(\alpha')) dG(\alpha') \right]^{n-1} d\pi > \int_{\pi \in \mathbb{R}} f(\pi) n [F(\pi)]^{n-1} d\pi = 1.$$

Proof of (v): By Lemma 3 (c), $\alpha = 0$ is the maximizer of $x^0(\alpha)$. Then, by Claim 3 (iv), $\sigma_0(0) > 1$.

Proof of (vi): Note that

$$\begin{aligned} \frac{\partial \sigma_{\hat{\alpha}}}{\partial \hat{\alpha}}(\alpha) &= \int_{\pi \in \mathbb{R}} f(\pi) \left[n(n-1) \left(\int_{\alpha' \in [0,1]} F(\pi + x^{\hat{\alpha}}(\alpha) - x^{\hat{\alpha}}(\alpha')) dG(\alpha') \right)^{n-2} \right. \\ &\quad \times \left. \int_{\alpha' \in [0,1]} f(\pi + x^{\hat{\alpha}}(\alpha) - x^{\hat{\alpha}}(\alpha')) \left(\frac{\partial x^{\hat{\alpha}}}{\partial \hat{\alpha}}(\alpha) - \frac{\partial x^{\hat{\alpha}}}{\partial \hat{\alpha}}(\alpha') \right) dG(\alpha') \right] d\pi. \end{aligned}$$

Since $\frac{\partial^2 x^{\hat{\alpha}}(\alpha)}{\partial \alpha \partial \hat{\alpha}} > 0$ by Lemma 3 (e), $\sigma_{\hat{\alpha}}(0)$ is decreasing in $\hat{\alpha}$, and $\sigma_{\hat{\alpha}}(1)$ is increasing in $\hat{\alpha}$.

Proof of (vii): Suppose that $\sigma_1(0) > 1$. Then, Claim 3 (vi) implies that $\sigma_{\hat{\alpha}}(0) > 1$ for each $\hat{\alpha} \leq 1$. Further, by Claim 3 (iii), $\sigma_{\hat{\alpha}}(1) < 1$.

As $x^{\hat{\alpha}}$ is single-peaked, $\sigma_{\hat{\alpha}}(0) > 1$ implies that there is α_0 such that for each $\alpha < \alpha_0$, $\sigma_{\hat{\alpha}}(\alpha) > 1$ and $\sigma_{\hat{\alpha}}(\alpha) \leq 1$ for each $\alpha \geq \alpha_0$. This is equivalent to that for each $\alpha < \alpha_0$, $\mathcal{G}'_{\hat{\alpha},G}(\alpha) > g(\alpha)$ and $\mathcal{G}'_{\hat{\alpha},G}(\alpha) \leq g(\alpha)$ for each $\alpha \geq \alpha_0$. In other words, G and $\mathcal{G}_{\hat{\alpha},G}$ are single-crossing. By Theorem 1.A.12 of Shaked and Shanthikumar (2007), $G \succ_{\text{fsd}} \mathcal{G}_{\hat{\alpha},G}$. $\square$

Now we proceed the proof of the theorem. For some α and α' , $\sigma_{\hat{\alpha}}(\alpha) > 1 > \sigma_{\hat{\alpha}}(\alpha')$ since $\int \sigma_{\hat{\alpha}}(\alpha) g(\alpha) d\alpha = \int \mathcal{G}'_{\hat{\alpha},G}(\alpha) d\alpha = 1$. Therefore, because $\sigma_{\hat{\alpha}}(\cdot)$ is single-peaked from Claim 3 (ii), exactly one of the following holds: Case (i) $\sigma_{\hat{\alpha}}(0) > 1 > \sigma_{\hat{\alpha}}(1)$. Case (ii) $1 > \max\{\sigma_{\hat{\alpha}}(0), \sigma_{\hat{\alpha}}(1)\}$. Case (iii) $\sigma_{\hat{\alpha}}(1) > 1 > \sigma_{\hat{\alpha}}(0)$.

In cases (i) and (iii), since $g(\alpha) > \mathcal{G}'_{\hat{\alpha},G}$ if and only if $\sigma_{\hat{\alpha}}(\alpha) < 1$, G and $\mathcal{G}_{\hat{\alpha},G}$ are single-crossing. Then, by Theorem 1.A.12 of Shaked and Shanthikumar (2007), $G \succ_{\text{fsd}} \mathcal{G}_{\hat{\alpha},G}$ in case (i), and $\mathcal{G}_{\hat{\alpha},G} \succ_{\text{fsd}} G$ in case (iii). In case (ii), we can show that $G \succ_{\text{uv}} \mathcal{G}_{\hat{\alpha},G}$ from the definition of the uniform variability order as $\sigma_{\hat{\alpha}}(\cdot) = \mathcal{G}'_{\hat{\alpha},G}(\cdot)/g(\cdot)$ is single-peaked and double crossing.

We now investigate which case is realized depending on the degree of $\hat{\alpha}$. First, by Claim 3 (v), $\sigma_0(0) > 1$. Therefore, if $\hat{\alpha} = 0$, case (a) holds.

Second, by Claim 3 (vi), all possible patterns of $\sigma_{\hat{\alpha}}(0)$ and $\sigma_{\hat{\alpha}}(1)$ are illustrated in Figure 2. Then, as Figure 2 illustrates, $\sigma_{\hat{\alpha}}(\cdot)$ is in case (i) for each $\hat{\alpha} < \hat{\alpha}_1$, in case (ii) for each $\hat{\alpha} \in (\hat{\alpha}_1, \hat{\alpha}_2)$, and in case (iii) for each $\hat{\alpha} > \hat{\alpha}_2$.

We note that $\hat{\alpha}_1 = \hat{\alpha}_2 < 1$ is impossible. If this is the case, as Figure 3 shows, both $\sigma_{\alpha}(1)$ and $\sigma_{\alpha}(0)$ are larger than 1. This is impossible by Claim 3 (iii).

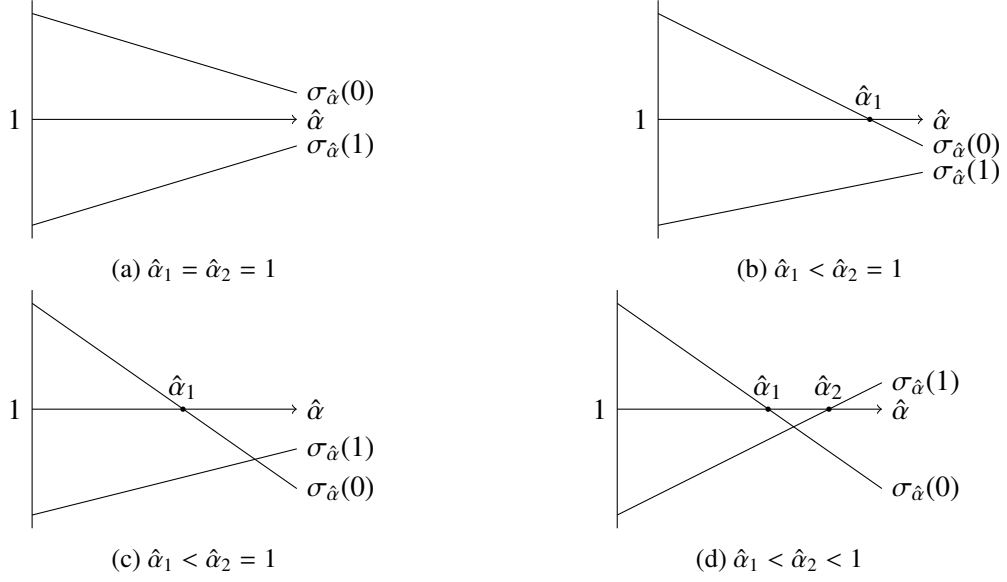


Figure 2: Patterns of the relations between $\sigma_{\hat{\alpha}}(0)$ and $\sigma_{\hat{\alpha}}(1)$

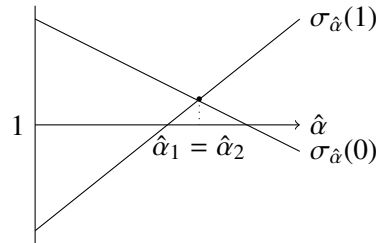


Figure 3: The case for $\hat{\alpha}_1 = \hat{\alpha}_2 < 1$

Finally, we show that if $x^1(0) > x^1(1)$, Case (iii) is impossible. By the definition of σ ,

$$\begin{aligned}\sigma_1(0) &= \int_{\pi \in \mathbb{R}} f(\pi) n \left[\int_{\alpha' \in [0,1]} F(\pi + x^1(0) - x^1(\alpha')) dG(\alpha') \right]^{n-1} d\pi \\ &> \int_{\pi \in \mathbb{R}} f(\pi) n \left[\int_{\alpha' \in [0,1]} F(\pi + x^1(1) - x^1(\alpha')) dG(\alpha') \right]^{n-1} d\pi = \sigma_1(1).\end{aligned}$$

By Claim 3 (vi), $\sigma_{\hat{\alpha}}(0)$ is decreasing in $\hat{\alpha}$ and $\sigma_{\hat{\alpha}}(1)$ is increasing in $\hat{\alpha}$. Therefore, $\sigma_{\hat{\alpha}}(0) \geq \sigma_1(0) > \sigma_1(1) \geq \sigma_{\hat{\alpha}}(1)$. This shows that case (iii) is impossible, showing that there is no $\hat{\alpha}$ such that $\mathcal{G}_{\hat{\alpha},G} \succ_{\text{fsd}} G$. $\square$

Proof of Theorem 3. Proof of (a): Let $z: [0, 1]^2 \rightarrow \{1, 0\}$ so that

$$z(\alpha, \hat{\alpha}) = \begin{cases} 1 & \text{if } \alpha = \arg \max_{\alpha' \in \text{supp}(G)} x^{\hat{\alpha}}(\alpha') \\ 0 & \text{otherwise.} \end{cases}$$

Then, we show that $\mathcal{G}_{\hat{\alpha},G}$ is independent of G other than its support in the limit.

Claim 4. Consider the limit $n \rightarrow \infty$. Then, $\mathcal{G}'_{\hat{\alpha},G}(\alpha) = \int z(\alpha, \hat{\alpha}) d\hat{G}(\hat{\alpha})$.

Proof of Claim 4. $\mathcal{G}'_{\hat{\alpha},G}(\alpha)$ is calculated as

$$\mathcal{G}'_{\hat{\alpha},G}(\alpha) = \int_{\hat{\alpha} \in [0,1]} \int_{\pi \in \mathbb{R}} \frac{f(\pi - x^{\hat{\alpha}}(\alpha))g(\alpha)}{\int f(\pi - x^{\hat{\alpha}}(\alpha'))dG(\alpha')} [(H(\pi | \hat{\alpha}, G))^n]' d\pi d\hat{G}(\hat{\alpha}).$$

Note that $[(H(\pi | \hat{\alpha}, G))^n]' = H'(\pi | \hat{\alpha}, G)n(H(\pi | \hat{\alpha}, G))^{n-1} \rightarrow 0$ as $n \rightarrow \infty$ if $\pi < \infty$. This implies that at the limit $n \rightarrow \infty$,

$$\mathcal{G}'_{\hat{\alpha},G}(\alpha) = \lim_{\pi \rightarrow \infty} \int_{\hat{\alpha} \in [0,1]} \lim_{\pi \rightarrow \infty} \frac{f(\pi - x^{\hat{\alpha}}(\alpha))g(\alpha)}{\int f(\pi - x^{\hat{\alpha}}(\alpha'))dG(\alpha')} d\hat{G}(\hat{\alpha}).$$

By the assumption that $\lim_{x \rightarrow \infty} f(x - a)/f(x - b) \rightarrow 0$ for any $a < b$,

$$\lim_{\pi \rightarrow \infty} \frac{f(\pi - x^{\hat{\alpha}}(\alpha))g(\alpha)}{\int f(\pi - x^{\hat{\alpha}}(\alpha'))dG(\alpha')} = \begin{cases} 1 & \text{if } \alpha = \arg \max_{\alpha'} x^{\hat{\alpha}}(\alpha') \\ 0 & \text{otherwise.} \end{cases}$$

This completes the proof. $\square$

By Lemma 3 (c) and (d), let $\chi(\hat{\alpha}) := \arg \max_{\alpha} x^{\hat{\alpha}}(\alpha)$, which is increasing in $\hat{\alpha}$ and less than $\hat{\alpha}$. Then, $\mathcal{G}_{\hat{\alpha},G}(\alpha) = \hat{G}(\chi^{-1}(\alpha)) \geq \hat{G}(\alpha)$.

Proof of (b): First note that by Theorem 3 (a), $\mathcal{G}_{\hat{G},G}(\alpha) = \hat{G}(\chi^{-1}(\alpha))$, where $\chi(\hat{\alpha}) := \arg \max_{\alpha} x^{\hat{\alpha}}(\alpha)$.

Next, we show that $\text{supp}(\mathcal{G}_{\hat{G},G})$ is an interval if $\text{supp} G$ is an interval. By the continuity of u and v , each e_s and e_m is continuous with respect to α and $\hat{\alpha}$. By the single-peakedness of $x^{\hat{\alpha}}(\cdot)$, $\arg \max_{\alpha' \in \text{supp}(G)} x^{\hat{\alpha}}(\alpha')$ is single-valued. Then, $\arg \max_{\alpha' \in \text{supp}(G)} x^{\hat{\alpha}}(\alpha')$ is also continuous in $\hat{\alpha}$. Note that when $\hat{\alpha} = \min \text{supp}(G)$, $\hat{\alpha} = \arg \max_{\alpha' \in \text{supp}(G)} x^{\hat{\alpha}}(\alpha')$. Then, the intermediate value theorem implies that for each $\hat{\alpha} \in \text{supp}(\mathcal{G}_{\hat{G},G})$ and each $\hat{\alpha}' < \hat{\alpha}$, $\hat{\alpha}' \in \text{supp}(\mathcal{G}_{\hat{G},G})$. This shows that $\text{supp}(\mathcal{G}_{\hat{G},G})$ is an interval.

Note also that $\max \text{supp}(\hat{G}_t)$ is a strictly decreasing sequence of real numbers. This is because, if $\text{supp}(\hat{G}_t) = [\underline{\alpha}_t, \bar{\alpha}_t]$, $\chi(\bar{\alpha}_t) < \bar{\alpha}_t$. Then, it converges. Let the limit be $\bar{\alpha}$. Suppose by contradiction that $\bar{\alpha} > \min \text{supp}(G)$. Then, $\bar{\alpha} > \arg \max_{\alpha' \in \text{supp}(G)} x^{\bar{\alpha}}(\alpha')$, which implies that $\bar{\alpha}$ is not in support of the manager's degree of bias. This contradicts that $\bar{\alpha}$ is the limit.

Proof of (c): Recall that the expected overall performance is single-peaked with respect to the manager's bias (Lemma 3, (f)). Let α^* be the maximizer of the expected overall performance. Theorem 3 (b) shows that the possible values of the manager's bias converge to the minimum value of $\text{supp}(G)$. Therefore, if t is sufficiently large, $\max \text{supp}(\mathcal{G}_{\hat{G}_t,G}) < \alpha^*$, and let t^* be the smallest t satisfying the inequality. After $t \geq t^*$, the manager with a larger bias among those with a positive measure yields greater overall performance. As $\hat{G}_t \succ_{\text{fsd}} \hat{G}_{t+1}$, generation after generation, the expected overall performance decreases. $\square$

Appendix B.4. Formal description and proof of Proposition 4

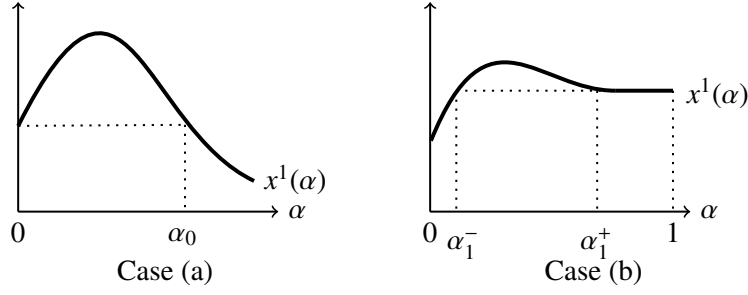
Proposition 4. (a) Suppose that $x^1(0) > x^1(1)$, and let α_0 be the solution to $x^1(0) = x^1(\alpha_0)$. Then, if $\int_0^{\alpha_0} g(\alpha) d\alpha$ is small enough, $G \succ_{\text{fsd}} \mathcal{G}_{\hat{\alpha},G}$ for each $\hat{\alpha}$.

(b) Suppose that $x^1(0) < x^1(1)$ and let α_1^- be the smallest solution to $x^1(1) = x^1(\alpha_1^-)$. Assume that $\min\{e_s^{\alpha_1^+}, e_m^{\alpha_1^+}\} = 1$ for some $\alpha_1^+ < 1$. Consider a distribution G that satisfies

$$\frac{\int_0^{\alpha_1^-} g(\alpha) d\alpha}{\int_{\alpha_1^+}^1 g(\alpha) d\alpha} \leq \mu \text{ and } \frac{\int_{\alpha_1^-}^{\alpha_1^+} g(\alpha) d\alpha}{\int_0^{\alpha_1^-} g(\alpha) d\alpha} \leq \mu.$$

Then, for each $\delta > 0$, if μ is sufficiently small, $\Pr_{\hat{\alpha} \sim G}(\{\hat{\alpha} \mid \mathcal{G}_{\hat{\alpha},G} \succ_{\text{fsd}} G\}) > 1 - \delta$.

To streamline the exposition, define the threshold $\alpha_0 > 0$ by $x^1(0) = x^1(\alpha_0)$ when $x^1(0) > x^1(1)$. When $x^1(0) < x^1(1)$ and some types choose corner solutions (i.e., $e_s^{\alpha} = e_m^{\alpha} = 1$), define thresholds $\alpha_1^- < \alpha_1^+ \in (0, 1)$ by $x^1(\alpha_1^-) = x^1(\alpha_1^+) = x^1(1)$. Figure 4 provides an illustration of these thresholds.



Each panel illustrates the shape of x^1 (i.e., the subordinate's performance when his manager's bias is $\hat{\alpha} = 1$). The left panel shows the graph of x^1 under the assumption of (a) and the right one shows that under (b).

Figure 4: Illustrations of $\alpha_0, \alpha_1^-, \alpha_1^+$ used in Proposition 4

Proof of Proposition 4. Proof of (a) Suppose that $x^1(0) > x^1(1)$. Then, we can find α_0 such that $x^1(0) = x^1(\alpha_0)$. By single-peakedness of x^1 , such α_0 is unique, and for each $\alpha > \alpha_0$, $x^1(0) > x^1(\alpha)$. Then, for each $\alpha > \alpha_0$, and each $\pi \in \mathbb{R}$, $F(\pi + x^1(0) - x^1(\alpha)) > F(\pi)$.

Therefore, if $\int_0^{\alpha_0} g(\alpha) d\alpha$ is small enough, $\int F(\pi + x^1(0) - x^1(\alpha')) dG(\alpha') > F(\pi)$. By using the same logic to prove Claim 3 (iv), we can show that $\sigma_1(0) > 1$. By Claim 3 (vii), $G \succ_{\text{fsd}} \mathcal{G}_{\hat{\alpha}, G}$ for each $\hat{\alpha}$. Thus, $G \succ_{\text{fsd}} \mathcal{G}_{G, G}$.

Proof of (b) Suppose that $x^1(0) < x^1(1)$. Then, we can find the smallest α_1^- such that $x^1(0) = x^1(\alpha_1^-)$. Then, by the single-peakedness of x^1 , $x^1(1) > x^1(\alpha)$ for each $\alpha \in [0, \alpha_1^-]$. Furthermore, by the assumption that $e_s^{\alpha_1^+} = 1$, and monotonicity of e_1^α , for each $\alpha \in [\alpha_1^+, 1]$, $e_1^\alpha = 1$. Then, as $x^{\hat{\alpha}}(\alpha) = v(e_s^\alpha, e_m^{\hat{\alpha}}) + u(e_s^{\hat{\alpha}})$, $x^1(\alpha) = x^1(1)$ for each $\alpha \in [\alpha_1^+, 1]$. Therefore, for each $\alpha \in [\alpha_1^+, 1]$ and $\alpha' \in [0, \alpha_1^-]$, $x^1(\alpha) > x^1(\alpha')$.

Hence, for each $\alpha \in [\alpha_1^+, 1]$,

$$\begin{aligned} F(\pi + x^1(1) - x^1(\alpha')) &= F(\pi) & \text{if } \alpha' \in [\alpha_1^+, 1], \\ F(\pi + x^1(1) - x^1(\alpha')) &> F(\pi) & \text{if } \alpha' \in [0, \alpha_1^-]. \end{aligned}$$

Then, if $\frac{\int_{\alpha_1^-}^{\alpha_1^+} g(\alpha) d\alpha}{\int_0^{\alpha_1^-} g(\alpha) d\alpha}$ is small enough, $\int F(\pi + x^1(1) - x^1(\alpha')) dG(\alpha') > F(\pi)$ and therefore $\sigma_1(1) > 1$.

Furthermore, for each $\hat{\alpha} > \alpha_1^+$, since $e_m^{\hat{\alpha}} = 1$, $\sigma_{\hat{\alpha}}(1) = \sigma_1(1) > 1$. As $\sigma_{\hat{\alpha}}$ is single-peaked, there is α^* such that for each $\sigma_{\hat{\alpha}}(\alpha) > 1$ if and only if $\alpha > \alpha^*$. By Claim 3 (i), $\mathcal{G}'_{\hat{\alpha}, G}$ and g are single-crossing and therefore, by Theorem 1.A.12 of Shaked and Shanthikumar (2007), $\mathcal{G}_{\hat{\alpha}, G} \succ_{\text{fsd}} G$ for each $\hat{\alpha} \in [\alpha_1^+, 1]$.

Hence, if $\frac{\int_0^{\alpha_1^-} g(\alpha) d\alpha}{\int_{\alpha_1^+}^1 g(\alpha) d\alpha}$ is small enough, $\Pr(\{\hat{\alpha} \mid \mathcal{G}_{\hat{\alpha}, G} \succ_{\text{fsd}} G\})$ can be arbitrary close to 1. $\square$

Supplementary Materials for Meritocratic Selection and Dynamics of Coordination Neglect in Hierarchical Organizations

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I. Numerical Analysis under effort complementarity

Section 5.2 shows that when the population is large enough, managerial bias decreases; however, overall performance declines over generations. By contrast, in a finite population, the evolution dynamics of managerial bias depend on the distribution of subordinates' biases. This section explores finite populations and numerically analyzes the impact of the performance-based promotion system on bias evolution and overall performance.

I.1. Numerical Analyses

We conduct a numerical analysis under the following specification:³⁰

$$\begin{aligned} v(e_s, e_m) &= -\frac{\eta}{2}(e_s)^2 + \eta_{ms}(1 - e_m)(1 - e_s) - \frac{\eta}{2}(e_m)^2 \\ u(e) &= \zeta(e - e^2/2) \end{aligned}$$

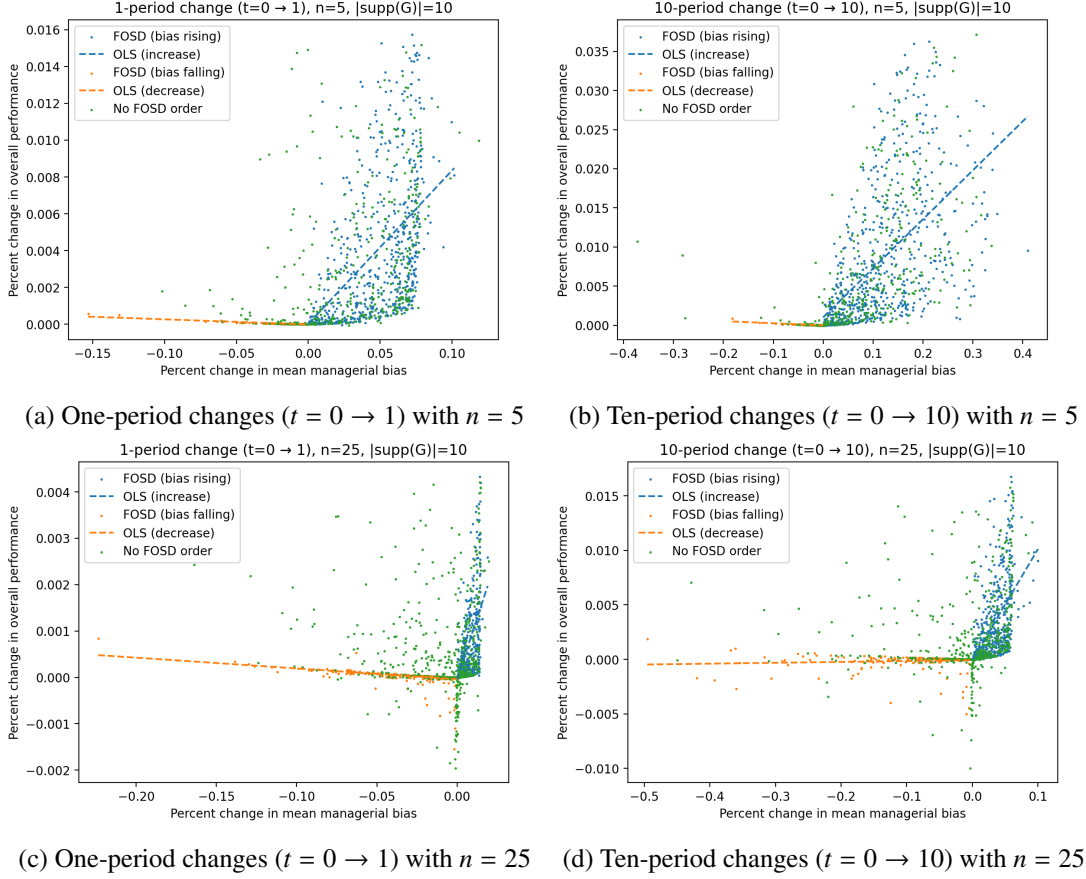
where η and ζ are positive and $\eta_{ms} > 0$.³¹ This is a simplified form of (1). We randomly draw 2000 distributions, each supported on 10 points in $[0.01, 1]$, as G , and compute the post-promotion managerial bias distribution $\mathcal{G}_{G,G}$ for each G , the distribution of bias among subordinates promoted when both the subordinate pool and the manager's bias follow G .

We first examine the relationship between changes in mean managerial bias and changes in overall performance. For each $H \in \{G, \mathcal{G}_{G,G}\}$, let $\alpha_H = \int \alpha dH(\alpha)$ and

$$\bar{\Pi}_H = \int \left[u(e_m^{\hat{\alpha}}) + \int \left(u(e_s^\alpha) + v(e_s^\alpha, e_m^{\hat{\alpha}}) \right) dG(\alpha) \right] dH(\hat{\alpha})$$

³⁰To facilitate exploration of the numerical results, we provide an interactive web application that reproduces the simulation environment used in the paper. The application allows users to vary key parameters and visualize the resulting dynamics of managerial bias and performance. It is available at <https://daido-tajika-meritocraticselection.onrender.com/>. The code is also available at https://github.com/Ttajika/Daido_Tajika_MeritocraticSelection

³¹In the current analysis, we set $n = 5$, $\eta = 0.05$, $\zeta = 1$, $\eta_{ms} = 1$, and F is the normal distribution with mean 0 and variance 0.01.



Each panel plots the relation in change rates between mean managerial bias and overall performance. The dashed lines show fitted regression lines. The blue line is estimated from observations in which the post-promotion bias distribution first-order stochastically dominates the original distribution, whereas the orange line is estimated from observations in which the opposite dominance relation holds.

Figure 5: Change rates in mean managerial bias and overall performance

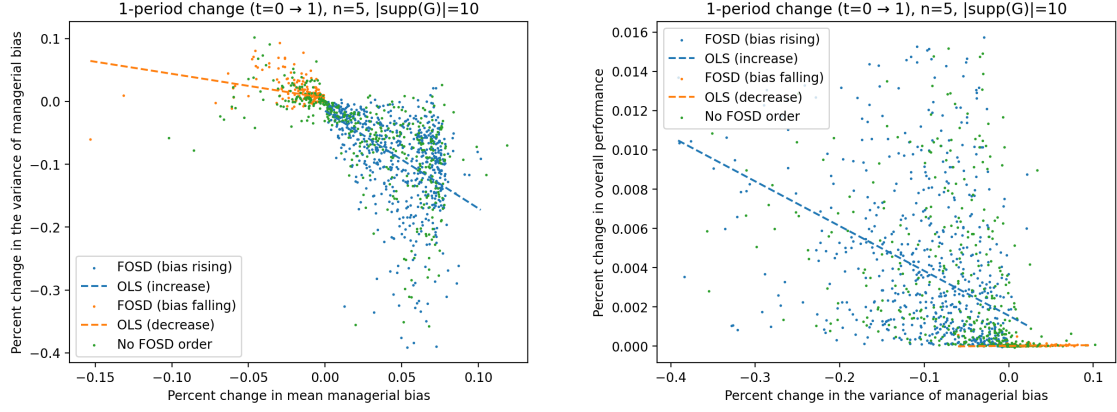
be the mean of distribution H and the expected overall performance when the manager's bias follows distribution H , respectively. We then compute the rate of change in overall performance, $(\bar{\Pi}_{\mathcal{G}_{G,G}} - \bar{\Pi}_G)/\bar{\Pi}_G$, and the rate of change in mean managerial bias, $(\alpha_{\mathcal{G}_{G,G}} - \alpha_G)/\alpha_G$. Figure 5(a) and (c) depict the results for $n = 5$ and 25. Figure 5(b) and (d) report the analogous comparison between $\hat{G}_{10}$ and G .³²

Next, we examine two additional relationships: the rate of change in mean bias versus the rate of change in variance, and the rate of change in variance versus the rate of change in performance. Figures 6(a) and 6(b) report the results.

These simulations yield three findings:

1. Overall performance generally improves after one round of promotion when $n = 5$, but performance declines become frequent when $n = 25$, especially over longer horizons ($t = 0 \rightarrow 10$). Performance gains tend to be larger when mean managerial bias increases (Figure 5).

³²Recall that $\hat{G}_t = \mathcal{G}_{\hat{G}_{t-1}, G}$ and $\hat{G}_0 = G$.



(a) Change rates in mean bias and variance of managerial bias (b) Change rates in variance of managerial bias and overall performance

Each panel uses the same color coding as Figure 5: the blue regression line corresponds to cases in which the new managerial-bias distribution first-order stochastically dominates the original one, and the orange line corresponds to the opposite relation.

Figure 6: The effects on variance and the relationship with performance and average bias

2. A larger increase in mean bias is associated with a larger reduction in the variance of bias (Figure 6 (a)).
3. A larger reduction in the variance of bias is associated with a larger improvement in performance (Figure 6 (b)).

These observations suggest that an increase in mean managerial bias is accompanied by a stronger reduction in the cross-sectional variance of bias, which in turn is associated with larger performance gains (Figure 6 (b)).

To provide a theoretical underpinning for this pattern, we use Theorem 2 to compare the distribution of the promoted subordinate's bias with the original subordinate distribution, conditional on the manager's bias $\hat{\alpha}$.

Recall that $\mathcal{G}_{\hat{\alpha},G}$ denotes the distribution of the promoted subordinate's bias when the manager's bias is fixed at $\hat{\alpha}$. Theorem 2 classifies the relationship between $\mathcal{G}_{\hat{\alpha},G}$ and G into three regions depending on $\hat{\alpha}$. Theorem 2 also demonstrates that if region (c) is non-empty, then region (b) is also non-empty, but the converse need not hold. When G stochastically dominates $\mathcal{G}_{G,G}$, implying a decrease in mean bias, only region (a) need be non-empty, and regions (b) and (c) may be empty. By contrast, when $\mathcal{G}_{G,G}$ stochastically dominates G , implying an increase in mean bias, region (c) is necessarily non-empty, so region (b) is non-empty as well, and the variance of bias tends to decrease. This explains the pattern in Figure 6 (a): when mean bias increases, the non-emptiness of regions (b) and (c) implies a compression of the bias distribution, which in turn improves performance.

In summary, when the subordinate pool is finite, meritocratic promotion can either increase or decrease mean managerial bias, depending on the initial distribution. Performance gains are larger when bias increases, because the associated variance compression identified in Theorem 2 regions (b) and (c) reinforces the benefits of selection.

Moreover, comparing the $n = 5$ and $n = 25$ panels of Figure 5, the performance decline predicted by the large-organization limit (Theorem 3 (c)) is already visible at moderate team sizes: at $n = 25$ and $t = 10$, a substantial fraction of initial distributions lead to negative performance changes, whereas such cases are rare at $n = 5$. Notably, these performance declines are concentrated among distributions for which mean bias decreases, while distributions with increasing bias continue to exhibit positive performance gains even at $n = 25$. This asymmetry is consistent with the mechanism underlying Theorem 3: it is the downward selection of managerial bias, rather than meritocratic selection per se, that drives the erosion of performance, and this effect emerges well before the large-organization limit.